

Computing the Random 3-SAT Threshold

OpenAI

September 27, 2026

Abstract

The limiting satisfiability threshold of uniform random 3-SAT is a computable real. We credit Gaia Carenini [4] with priority for resolving the satisfiability conjecture, which establishes the threshold’s existence. We prove that one finite deterministic machine can approximate it to any prescribed accuracy. A deletion estimate gives explicit lower certificates, while a finite hierarchical approximation of the soft pressure gives upper certificates at every larger rational density. A fair search through these certificates halts without requiring a computable rate of finite-size convergence.

1 Introduction

For $n \geq 3$, a proper random 3-clause chooses three distinct variables uniformly from $x_1, \dots, x_n$ and gives each variable an independent fair sign. Let $\Phi(n, m)$ be the conjunction of $m \geq 0$ independent clauses with this law. Thus its clauses are sampled with replacement from the $8\binom{n}{3}$ possibilities; the formula with no clauses is satisfiable. Write

$$p(n, m) = \mathbb{P}\{\Phi(n, m) \text{ is satisfiable}\}.$$

The ratio m/n is the clause density. We prove that the limiting density separating satisfiable and unsatisfiable formulas can be approximated to any prescribed accuracy by one finite algorithm.

Carenini [4, Corollary 1.2] establishes the existence of a limiting satisfiability threshold for every fixed $k \geq 3$. We credit Carenini with priority for the resolution of the satisfiability conjecture. The contribution here is computability of the 3-SAT threshold, beyond its existence.

Theorem 1.1. *There is a real number $\alpha_3 \in (0, \infty)$ such that, for every fixed real $a \geq 0$,*

$$\lim_{n \rightarrow \infty} p(n, \lfloor an \rfloor) = \begin{cases} 1, & a < \alpha_3, \\ 0, & a > \alpha_3. \end{cases}$$

There is one finite deterministic Turing machine which, on input 1^r for any integer $r \geq 1$, halts with a rational number q_r satisfying $|q_r - \alpha_3| \leq 2^{-r}$. Its only input is the unary precision 1^r ; it uses no oracle, advice, or noncomputable real constant.

The strict-side limits determine α_3 uniquely, since two different constants would give opposite limits at a density between them. The theorem leaves the behavior at $a = \alpha_3$ open. Its algorithm computes the limiting density; it does not search for satisfying assignments to an input formula. We obtain no efficiency bound, and the computability argument concerns clause size three.

Random 3-SAT has long served as a test case for the typical behavior of satisfiability problems. Experiments of Mitchell, Selman, and Levesque [18] placed difficult instances for the Davis–Putnam

procedure near the density where half the formulas were satisfiable. Algorithmic and counting arguments then constrained the transition’s location. Kaporis, Kirousis, and Lalas [14] obtained the lower bound 3.52, independently reported by Hajiaghayi and Sorkin [12]; Díaz, Kirousis, Mitsche, and Pérez-Giménez [6] obtained the upper bound 4.4898. These bounds bracket any limiting threshold. From statistical physics, Mézard, Parisi, and Zecchina developed the cavity and survey-propagation description [16]. The calculations of Mézard and Zecchina [17], refined by Mertens, Mézard, and Zecchina [15], predict a threshold near 4.267. Theorem 1.1 establishes existence and computability without evaluating the constant or identifying it with that prediction.

Two further distinctions explain the mathematical task. A transition window can become narrow while its center still varies with the system size. Friedgut’s theorem, with Bourgain’s appendix [9], establishes a sharp transition for every fixed $k \geq 3$ about a size-dependent location. In earlier work, Carenini [5, Corollaries 7.10–7.11] obtained an $O(n/\log n)$ central clause window for each fixed $k \geq 3$, including the model with independently sampled proper clauses. Her polynomial improvement to $O_{k,\eta}(n^{1/2+1/k})$ between probability levels η and $1 - \eta$, for each fixed $0 < \eta < 1/2$, in [4, Theorem 1.1] yields limiting thresholds for every fixed $k \geq 3$. Bayati, Gamarnik, and Tetali [3] developed sparse interpolation and proved limits for normalized optimization values and finite-temperature normalized log partition functions. Their near-satisfiability conclusion permits $o(n)$ violated clauses; exact satisfiability requires none. Ding, Sly, and Sun [7] proved the limiting-threshold conjecture, including the predicted value, for all sufficiently large fixed clause sizes. The probability argument below also proves existence at clause size three as part of the development of the computability result.

Even when a finite-size limit exists, it need not be a computable real. Specker [30, Satz IV] constructed a bounded nondecreasing computable sequence of rationals with a noncomputable limit. Our proof supplies two families of finite certificates. The first certifies rational lower bounds on α_3 that approach it arbitrarily closely. The second certifies rational upper bounds, with a certificate at every rational density strictly above α_3 . A fair search through both families produces certified intervals of arbitrarily small length. Its stopping rule uses the certified interval itself and requires no convergence modulus for the finite-size quantities.

2 Two certificate searches give a computable real

The following elementary lemma isolates the effective part of the proof. Its test value $G(a, \beta; Q)$ is an arbitrary computable real satisfying the two stated sign conditions. The random-formula construction will supply those conditions; the search itself uses only finite strings and certified rational approximations.

Lemma 2.1 (Finite-certificate search). *Let $\alpha \in [0, 20]$. Suppose that $(\ell_k)_{k \geq 1}$ is a computable sequence of rational numbers satisfying $\ell_k \leq \alpha$ for every k and $\ell_k \rightarrow \alpha$. Suppose also that finite descriptions Q form an effectively enumerable class, and that for every positive rational a , positive integer β , and valid description Q , the real number $G(a, \beta; Q)$ is uniformly computable. Assume that*

1. $G(a, \beta; Q) < 0$ implies $\alpha \leq a$;
2. for every rational $a > \alpha$, some positive integer β and valid description Q satisfy $G(a, \beta; Q) < 0$.

Then one finite deterministic Turing machine, with only the unary input 1^r for $r \geq 1$, returns a rational number within 2^{-r} of α . It can encode the output as a binary integer numerator and a positive integer denominator, with separators.

Proof. Uniform computability gives, from (a, β, Q) and an accuracy index, a rational closed interval containing $G(a, \beta; Q)$ whose width tends to zero as the index increases. If $G(a, \beta; Q) < 0$, one such

interval has a strictly negative upper endpoint. This semidecides strict negativity without testing equality to zero.

Start with $[l, u] = [0, 20]$. Fix an effective enumerator of the triples (a, β, Q) in the statement. At stage s , run it for s computation steps from its start. For every triple emitted during those steps, compute its value intervals at all accuracy indices at most s , and compute $\ell_1, \dots, \ell_s$. This is a finite list of terminating computations. Replace l by the maximum of its old value and the lower certificates just computed. Whenever a value interval has a strictly negative upper endpoint, replace u by $\min\{u, a\}$. After each update, if $u - l \leq 2^{1-r}$, output $(u + l)/2$ and halt.

Every interval maintained by the procedure contains α . Its lower endpoint does so by the hypothesis on ℓ_k , and its upper endpoint does so by assumption (1). To prove termination, fix $\varepsilon > 0$. Some $\ell_k > \alpha - \varepsilon$ is eventually computed. Choose a positive rational a with $\alpha < a < \alpha + \varepsilon$. By assumption (2), a particular triple at this density has a negative value. The enumerator emits this triple after finitely many steps, and some finite accuracy index certifies its negativity. Both are processed at a finite stage. Thereafter $l > \alpha - \varepsilon$ and $u < \alpha + \varepsilon$. The interval length tends to zero, so the stopping test succeeds for every r . Its midpoint has error at most 2^{-r} .

All operations in this schedule act on finite integer and rational encodings. The algorithms supplied by the hypotheses are fixed parts of one program. The density and witness chosen in the termination argument are not advice given to that program. $\square$

3 Where the certificates come from

The probability argument first constructs a threshold α for the same proper clause law as in Theorem 1.1. It therefore equals the $k = 3$ constant of *A Limiting Satisfiability Threshold for Every Fixed Clause Size* [19, Theorem 1.1]: different constants would force opposite limits at a density between them. The comparison identifies the constants. We give the probability argument locally because the computation needs two additional conclusions: explicit lower certificates and a positive density of unavoidable violations above the threshold.

Lower certificates and the sign of pressure. For this part of the proof, clauses arrive in a rate-one Poisson process and their three variable indices are sampled independently, so indices may repeat within a clause. Let T_n be the first unsatisfiable arrival time and put $f_n = \mathbb{E} \min\{T_n, 20n\}$. Section 4 proves

$$\frac{f_n}{n} \longrightarrow \alpha, \quad \frac{f_n}{n} - 2^{67} n^{-1/6} \leq \alpha.$$

Each f_n admits certified rational approximations by finite clause enumeration and integration. These facts give a computable sequence of rational lower certificates converging to α . Thinning and a comparison with fixed density slack transfer the threshold from the auxiliary law to proper clauses.

The estimate behind these conclusions bounds the extra survival time when one variable may be deleted for free. Its $6/5$ moment is bounded uniformly even when other deletions are already allowed. The replacement step follows the residual-set conditioning principle of Friedgut's half-cube lemma [9, Lemma 5.7] and Carenini's sequential clause replacement [5, Theorem 4.3 and Lemma 6.1]. Here independent variable positions give exact powers for two-literal and three-literal killing probabilities. A weighted interpolation between unequal blocks, followed by a weight-balancing argument, turns the moment bound into a summable error in the normalized mean. Keeping the existing deletion allowance has a second use: above α , every assignment violates a positive multiple of n clauses with probability tending to one.

For the auxiliary Poisson formula at time an , let $H_n(\sigma)$ count the clauses violated by an assignment σ . For a finite penalty $\beta > 0$, define

$$Z_n(a, \beta) = \sum_{\sigma \in \{0,1\}^n} e^{-\beta H_n(\sigma)}, \quad P(a, \beta) = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \log Z_n(a, \beta).$$

Section 5 proves that this pressure limit exists. Below α it is nonnegative for every $\beta > 0$. Above α , the linear violation bound makes it strictly negative for some positive integer β .

Finite upper certificates. The remaining task is to detect negative pressure using finite data. Section 5 defines an effectively enumerable family of rational trial descriptions. A description consists of finitely many rational averaging parameters and a finite table of rational pairs in $[0, 1]^2$. The two coordinates give messages for the two literal signs; they need not sum to one. Independent uniform coordinates select table entries through finitely many conditional sampling levels. Different sites may share initial random data, called the root data, but their subsequent coordinates are independent. The value $G(a, \beta; Q)$ of a trial is uniformly computable, and we prove

$$P(a, \beta) \leq G(a, \beta; Q), \quad P(a, \beta) = \inf_Q G(a, \beta; Q).$$

A certified negative value is therefore an upper certificate for α , and the infimum identity supplies one at every rational density above α .

The inequality and the infimum identity require different arguments. For the inequality, the three-literal interpolation leaves the nonnegative remainder $(x - y)^2(x + 2y)$ for $x, y \geq 0$. For the reverse approximation, finite formulas produce nearly optimal trials, but their later sampling levels can still share random variables between sites. Removing those variables is the main structural task. Conditional changes of sampling law and Gaussian tests describe the continuation law of a fresh site for almost every fixed history of the shared coordinates. This qualification lets the description survive later changes of law.

Section 6 first proves that changing all averaging parameters has a cost independent of the number of levels. This lets the tolerances be chosen before the number of structural reductions is known. Section 7 then obtains the conditional descriptions needed by those reductions. Each reduction either reaches the root or decreases the first remaining parameter by a fixed factor. After finitely many steps, Section 8 samples the retained conditional laws independently at each site, controls the change of the trial value, and approximates the resulting kernels by finite rational tables.

Finite hierarchical trials developed in the diluted interpolation and cavity tradition of Franz and Leone [8], Panchenko and Talagrand [27], and Aizenman, Sims, and Starr [1]. The upper-bound and reservoir calculations below use their interpolation and cavity mechanisms in the present three-literal model. Panchenko's ultrametricity theorem and the hierarchical representation of Austin and Panchenko [23, 2] supply the geometry and representation of the limiting random arrays. Panchenko's pure-state analysis [25] motivates the identities that also test the retained site data. The conditional-law records then control replacement of shared nonroot coordinates by independent site sampling. This approximation, followed by rational discretization, gives the finite trial completeness needed for the certificate search.

The proof may select subsequences and conditional laws non-effectively. Those selections establish that a finite witness exists. The algorithm searches finite descriptions; it does not receive a selected law or a convergence modulus. Section 9 closes the argument by checking the two hypotheses of Lemma 2.1.

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4 Probability estimates for the two certificate searches

The certificate search needs two facts about random formulas. First, it needs computable lower bounds that approach the limiting density. Second, it needs a nonnegative lower limit below the threshold and a negative upper limit above it for normalized expected log partition functions, with a suitable finite penalty in the latter case. These become pressure signs when the pressure limit is established. The common estimate controls the extra time for which a formula stays satisfiable when one variable may be deleted, even if other deletions are already allowed.

It is useful to begin with an auxiliary clause law. For every integer $n \geq 1$, clauses arrive in a Poisson process of rate one. Each of their three variable indices is sampled independently and uniformly from $[n]$, and each sign is sampled independently and fairly. All indices, signs, and the clock are independent. Repetitions *within* a clause are permitted; time *an* therefore gives a Poisson number of these clauses with mean an . We call this the *relaxed Poisson model*. Later in the section we transfer its strict-side limits to the proper-clause model of the introduction.

Fix throughout this section

$$B = 20, \quad p = \frac{6}{5}, \quad b = \frac{1}{p} = \frac{5}{6}. \quad (4.1)$$

For an integer r with $0 \leq r \leq n$, a formula is called r -deletion satisfiable if some set of at most r variables can be deleted so that the remaining formula is satisfiable; deleting a variable discards every clause that touches it. Let T_r be the first time r -deletion satisfiability fails, with $T_r = +\infty$ if it never fails, and put

$$F_r = T_r \wedge Bn.$$

The formula at time t includes all arrivals up to and including t . Accordingly it survives at time t exactly when $T_r > t$. For a fixed variable v , a superscript $+v$ means that v is deleted for free and up to r *other* variables may be deleted. In particular, $F_r^{+v} = T_r^{+v} \wedge Bn \geq F_r$, with the same cap Bn . Allowing more deletions than there are other variables has its literal meaning; it simply permits deleting all of them. We suppress the dependence on n in these times. For this fixed size, the

overview's T_n is T_0 here; both denote the uncapped first-unsatisfiable time, whereas $F_0 = T_0 \wedge Bn$ is capped. Write

$$f_n = \mathbb{E}F_0, \quad P_n(t) = \mathbb{P}(T_0 > t).$$

Thus f_n is the mean first-unsatisfiable time capped at $20n$.

4.1 A uniform moment bound for one free deletion

Lemma 4.1 (Free-deletion moment). *For all integers $n \geq 1$ and $0 \leq r \leq n$, and every $v \in [n]$,*

$$\mathbb{E}[(F_r^{+v} - F_r)^p] \leq C, \quad C = 2^{60}. \quad (4.2)$$

Consequently, uniformly in r ,

$$\mathbb{E}|F_r - \mathbb{E}F_r|^p \leq C_1 n, \quad C_1 = 2^{62}. \quad (4.3)$$

The proof compares two ways to eliminate a set of surviving assignments. Clauses containing the free variable once become two-literal tests after that variable is fixed. A batch of relaxed three-literal tests can replace each such test at controlled loss. This is the residual-set conditioning principle used in Friedgut's half-cube lemma [9, Lemma 5.7] and in Carenini's sequential clause replacement [5, Theorem 4.3 and Lemma 6.1]. The calculation below is for independently sampled positions: their exact powers give the estimate needed under an existing deletion allowance.

Proof. For $n = 1$, Equation (4.2) follows directly from $0 \leq F_r^{+v} - F_r \leq 20$. Suppose henceforth that $n \geq 2$.

Incident clauses. Split the Poisson process into the clauses avoiding v , called the *base*, and the clauses involving v . The base has rate

$$\lambda_n = \left(\frac{n-1}{n}\right)^3 \geq \frac{1}{8},$$

and its clauses have the relaxed distribution on the other $n-1$ variables. It is independent of the incident streams.

At a deterministic time t , condition on the entire base up to t . Let U_t be its finite set of surviving pairs (D, σ) , where $D \subseteq [n] \setminus \{v\}$ has size at most r and σ is an assignment on $[n] \setminus (D \cup \{v\})$. When a literal is tested on such a pair, a literal on D is declared true. This convention exactly implements clause deletion. The event $U_t \neq \emptyset$ is $\{T_r^{+v} > t\}$. For a nonempty deterministic candidate set U , let $q(U)$ be the probability that $\text{Poi}(K)$ independent relaxed two-literal clauses on the other variables eliminate every candidate, where $K = 3B = 60$.

For $0 \leq t < Bn$ and $U_t \neq \emptyset$, we claim

$$\mathbb{P}(T_r \leq t \mid \text{base up to } t) \leq (q(U_t) + 60/n)^2. \quad (4.4)$$

To check this, fix a value for v . Only incident clauses in which every occurrence of v has the sign falsified by that value can obstruct an extension of a candidate in U_t . Among these clauses, those containing v once project to independent relaxed two-literal clauses. Their number is Poisson with mean at most $3t/n \leq 60$. The expected number containing v at least twice is at most $3t/n^2 \leq 60/n$, by a union bound over the three pairs of positions. The obstruction probability for this value is therefore at most $q(U_t) + 60/n$. The obstructing groups for the two values of v are disjoint Poisson thinnings, hence independent of one another and of the base. If either group leaves a candidate, restoring v with that value gives an r -deletion satisfying assignment. Both groups must therefore obstruct, which proves Equation (4.4).

Replacing shorter tests. In any list of independent tests, replacing one relaxed two-literal clause by $d \geq 1$ independent relaxed three-literal clauses reduces the probability of eliminating all candidates by at most d^{-2} . Indeed, condition on all other tests, leaving a candidate set U' . There is nothing to prove if it is empty. Otherwise let x be the proportion of signed literals false on every member of U' . Each variable contributes at most one such sign, so $0 \leq x \leq 1/2$, including under the deleted-literal convention. The two-literal clause alone eliminates U' with probability x^2 . The d three-literal clauses do so with probability at least $1 - (1 - x^3)^d \geq 1 - e^{-dx^3}$. If $dx^3 > 1$, this lower bound exceeds $1/4 \geq x^2$. If $dx^3 \leq 1$, it is at least $dx^3/2$, and

$$x^2 - \frac{d}{2}x^3 \leq \frac{16}{27d^2} < \frac{1}{d^2}.$$

This proves the comparison. Applying it successively is valid because, at each replacement, one can condition on all the other independent tests.

How long a residual set can survive. The replacement estimate converts a large value of $q(U_t)$ into a chance that the future base eliminates U_t . Partition the possible values $q(U_t) \geq 1/n$ into disjoint dyadic bins $q_0 \leq q(U_t) \leq 2q_0$, with $q_0 = 2^{-j}$ and $j \geq 1$, using any fixed endpoint convention. Every bin that is used has $q_0 \geq 1/(2n)$. For its index j , set

$$L_j = j + 122, \quad d_j = \left\lceil \sqrt{4L_j/q_0} \right\rceil, \quad \ell_j = 32L_j d_j. \quad (4.5)$$

Since $\mathbb{E}2^{\text{Poi}(60)} = e^{60} < 2^{120}$, $\mathbb{P}(\text{Poi}(60) > L_j) \leq q_0/4$. If U belongs to the bin, then L_j two-literal tests eliminate it with probability at least $q(U) - q_0/4$. Replacing each test by d_j three-literal tests loses at most $L_j/d_j^2 \leq q_0/4$. Thus $L_j d_j$ three-literal tests eliminate U with probability at least $q_0/2$.

In a future interval of length ℓ_j , the base has a Poisson number of arrivals with mean at least $4L_j d_j$. If $M = L_j d_j \geq 1$ and X is Poisson with mean $\mu \geq 4M$, Chebyshev's inequality gives

$$\mathbb{P}(X < M) \leq \frac{\mu}{(\mu - M)^2} \leq \frac{4}{9M} < \frac{1}{2}.$$

Consequently this interval eliminates the fixed U with probability at least $q_0/4$. Write

$$S_{r,v}(t) = \mathbb{P}(T_r^{+v} > t), \quad E_j(t) = \{U_t \neq \emptyset, q(U_t) \text{ belongs to bin } j\}.$$

The independent future base then gives

$$S_{r,v}(t) - S_{r,v}(t + \ell_j) \geq (q_0/4)\mathbb{P}(E_j(t)), \quad \int_0^{Bn} \mathbb{P}(E_j(t)) \, dt \leq 4\ell_j/q_0. \quad (4.6)$$

For the second inequality, integrate the first: the integral of $S_{r,v}(t) - S_{r,v}(t + \ell_j)$ over $[0, Bn]$ is at most ℓ_j . This argument does not require T_r^{+v} to be finite. Conditional on the base at time t in $E_j(t)$, survival to $t + u$ is also bounded by

$$(1 - q_0/4)^{\lfloor u/\ell_j \rfloor}. \quad (4.7)$$

For each complete future interval, failure to eliminate even the original U_t is necessary for survival; these interval events are independent.

Integrating the delay. For any $0 \leq F_r \leq F_r^{+v} \leq Bn$, direct integration gives the pathwise identity

$$(F_r^{+v} - F_r)^p = p(p-1) \int_{\substack{t, u \geq 0 \\ t+u < Bn}} u^{p-2} \mathbf{1}_{\{T_r \leq t, T_r^{+v} > t+u\}} \, dt \, du. \quad (4.8)$$

The boundaries have Lebesgue measure zero, so a hitting time equal to the cap causes no difficulty. Conditional on the base up to t , incident arrivals up to t and future base arrivals are independent. The incident killing estimate (4.4) and the future survival estimate (4.7) therefore multiply.

Here is why the resulting sum over bins converges. On a bin of scale q_0 , incident killing has probability at most a constant times q_0^2 . The expected total time spent in that bin before Bn is at most $4\ell_j/q_0$, by Equation (4.6). Integrating the geometric future-survival bound against u^{p-2} contributes at most a constant times $(\ell_j/q_0)^{p-1}$. Thus the bin contribution is bounded by a constant times $\ell_j^p q_0^{2-p}$. Since ℓ_j grows like a polynomial in j times $q_0^{-1/2}$, the power of q_0 is $2 - 3p/2 = 1/5 > 0$.

We now retain explicit constants for the later one-sided certificate. On bin j ,

$$(q(U_t) + 60/n)^2 \leq (122q_0)^2 < 2^{14}q_0^2, \quad \ell_j \leq 2^{20}(j+1)^{3/2}2^{j/2}.$$

For the latter, $d_j \leq 3\sqrt{L_j/q_0}$ and $L_j \leq 64(j+1)$ suffice. Moreover,

$$\begin{aligned} \int_0^\infty u^{p-2}(1 - q_0/4)^{\lfloor u/\ell_j \rfloor} du \\ \leq e \Gamma(1/5)(4\ell_j/q_0)^{1/5} < 2^7(\ell_j/q_0)^{p-1}. \end{aligned}$$

Indeed, the geometric term is at most $e \exp(-q_0 u/(4\ell_j))$, and splitting the defining Gamma integral at 1 gives $\Gamma(1/5) < 6$. Using Equation (4.6) and $p(p-1) < 1$, the contribution of bin j to the expectation in Equation (4.8) is at most

$$2^{47}(j+1)^2 2^{-j/5}.$$

Grouping $j = 5k + 1, \dots, 5k + 5$ gives

$$\sum_{j \geq 1} (j+1)^2 2^{-j/5} \leq 5 \sum_{k \geq 0} (5k+6)^2 2^{-k} = 1710 < 2^{11}.$$

The total bin contribution is less than 2^{58} . When $q(U_t) < 1/n$, Equation (4.4) is at most $61^2/n^2$. Integrating over the entire truncated region in Equation (4.8) bounds the remaining contribution by

$$61^2 B^p n^{p-2} < 2^{18}.$$

This proves Equation (4.2) with $C = 2^{60}$, without assuming that the free-deletion problem eventually fails. In particular, if $r \geq n-1$, a candidate can delete all other variables, so $q(U_t) = 0$ and only the small- q estimate is used.

From deletion to concentration. Partition all arrivals up to Bn into n independent buckets according to the smallest variable index in a clause. Resample bucket i independently, and denote the resulting capped time by F'_r . Deleting i removes that entire bucket, so F_r^{+i} is unchanged by the resampling and dominates both F_r and F'_r . Hence

$$\mathbb{E}|F_r - F'_r|^p \leq \mathbb{E}(F_r^{+i} - F_r)^p + \mathbb{E}(F_r^{+i} - F'_r)^p \leq 2C.$$

Reveal the buckets in their index order and let D_i be the successive differences of the Doob martingale for F_r . Averaging over the independent replacement and all unrevealed buckets identifies D_i as the conditional expectation of $F_r - F'_r$ given the first i buckets. Conditional Jensen therefore gives $\mathbb{E}|D_i|^p \leq 2C$.

For real x, y and $1 < p < 2$,

$$|x + y|^p \leq |x|^p + p \operatorname{sgn}(x)|x|^{p-1}y + 2|y|^p.$$

Here $\operatorname{sgn}(0) = 0$. The signed $(p-1)$ -power is Hölder continuous with constant 2^{2-p} , and integration of the derivative gives a remainder at most $2^{2-p}|y|^p \leq 2|y|^p$. Apply this inequality successively to the centered martingale sum. Each linear term has expectation zero, giving $\mathbb{E}|F_r - \mathbb{E}F_r|^p \leq 2 \sum_i \mathbb{E}|D_i|^p \leq 4Cn$. This is Equation (4.3). $\square$

4.2 Unequal blocks and a summable merging error

The moment bound places the capped time close to its mean on a scale $n^{5/6}$. To make the normalized means converge, we compare two independent blocks with their union. The times assigned to the two blocks may be arbitrary, so their densities need not agree.

Lemma 4.2 (Weighted satisfiability interpolation). *For positive integers n_1, n_2 and real $t_1, t_2 \geq 0$,*

$$P_{n_1+n_2}(t_1 + t_2) \geq P_{n_1}(t_1)P_{n_2}(t_2). \quad (4.9)$$

Proof. Mixing the blocks. Put $n = n_1 + n_2$, $t = t_1 + t_2$. The case $t = 0$ is immediate, so set $\lambda_i = t_i/t$. A *local* clause first chooses block i with probability λ_i and then chooses its three indices uniformly in that block. A *weighted global* clause chooses its block independently at each of the three positions with these same probabilities, followed by a uniform index in the chosen block. Interpolate between independent Poisson counts of local and weighted global clauses, with respective means $(1-s)t$ and st , for $0 \leq s \leq 1$.

The derivative of the expected satisfiability indicator is t times the expected difference of its add-one increments for the two clause types. This follows by differentiating the Poisson series for a bounded function; the differentiated series is absolutely convergent. For a satisfiable base formula, let x_i be the proportion of signed literals in block i that are false in every satisfying assignment. A single clause eliminates all satisfying assignments exactly when each of its literals is false in every such assignment. The local and weighted global killing probabilities are therefore

$$\sum_{i=1}^2 \lambda_i x_i^3, \quad \left(\sum_{i=1}^2 \lambda_i x_i \right)^3,$$

respectively. Convexity of x^3 on $[0, \infty)$ makes the derivative nonnegative. An unsatisfiable base contributes zero. At the local endpoint the two blocks are independent, so its satisfiability probability is the right-hand side of Equation (4.9).

Balancing the variable weights. The mixed endpoint still has weight λ_i/n_i on each variable of block i . It remains to show that making these weights uniform cannot decrease the probability at the weighted global endpoint. We prove this for every fixed clause count. Choose two variables with weights w_1, w_2 and replace both weights by $(w_1 + w_2)/2$. If their sum is zero, there is nothing to change. Otherwise condition on the set of occurrence positions that use one of these two variables, and on all choices and signs in the other positions. These conditioned data have the same distribution before and after averaging. Let k be the number of selected positions. Treating their literal truth values as independent placeholders defines a set $S \subseteq \{0, 1\}^k$: a vector belongs to S if the variables outside the selected pair can be assigned so that the formula is satisfied. This set does not depend on the identities or signs still to be sampled in the selected positions.

Let y be the vector of literal truth values when the two selected variables are both assigned zero, and let ξ indicate positions occupied by the second variable. Fair independent signs make y uniform on the bit cube, independently of ξ , whose coordinates are independent Bernoulli variables with parameter $\theta = w_2/(w_1 + w_2)$. The four assignments to the selected pair yield exactly the truth vectors

$$y, \quad \bar{y}, \quad y \oplus \xi, \quad \overline{y \oplus \xi}.$$

Here a bar denotes coordinatewise complement. Let $A = S \cup \{\bar{z} : z \in S\}$ and $g = \mathbf{1}_A$. The conditional probability of satisfaction is consequently

$$2\mathbb{E}g(y) - \mathbb{E}[g(y)g(y \oplus \xi)].$$

Expand g in the orthonormal parity characters $\chi_I(y) = (-1)^{\sum_{j \in I} y_j}$ of the uniform bit cube. Independence of the selector bits gives

$$\mathbb{E}[g(y)g(y \oplus \xi)] = \sum_{I \subseteq [k]} \hat{g}(I)^2 (1 - 2\theta)^{|I|}.$$

Complement invariance of g makes every odd-degree coefficient vanish. Every remaining nonconstant term is nonnegative and is zero at $\theta = 1/2$. Thus equalizing the selected weights maximizes the conditional satisfiability probability.

Repeatedly average a largest and a smallest weight. The sum over all weights of squared deviations from $1/n$ decreases by half the square of their difference. This nonnegative sum converges, so the largest-smallest difference tends to zero and all weights converge to $1/n$. For a fixed clause count the satisfiability probability is a polynomial in the weights, hence continuous. It cannot decrease in this averaging procedure. Bounded convergence then permits averaging over the Poisson count. The uniform endpoint is the left-hand side of Equation (4.9), proving the result. $\square$

Proposition 4.3 (A limiting center and a one-sided error). *The sequence f_n/n converges to a real number α . For every integer $k \geq 1$,*

$$\alpha \geq \frac{f_k}{k} - 2^{67} k^{-1/6}. \quad (4.10)$$

Proof. We first turn the survival comparison into an almost-superadditive inequality for f_n . Then a dyadic construction makes its accumulated error summable after division by the size. Set $H = 2^{60}$ and $h_k = Hk^b$. Equation (4.3) and Markov's inequality give

$$\mathbb{P}(|F_0 - f_k| \geq h_k) \leq C_1/H^p = 2^{-10} < 1/10$$

at size k . In Lemma 4.2, choose $t_i = (f_{n_i} - h_{n_i})_+$. If $t_i = 0$, survival is certain. Otherwise $t_i < Bn_i$, so survival has probability at least $9/10$ by this deviation bound. With $n = n_1 + n_2$ and $t = t_1 + t_2$, Equation (4.9) gives $P_n(t) \geq 81/100$. Also $t < Bn$. If $f_n < t - h_n$, the same deviation bound at size n would give $P_n(t) \leq 1/10$, a contradiction. Hence

$$f_{n_1+n_2} \geq f_{n_1} + f_{n_2} - C_2(n_1 + n_2)^b, \quad C_2 = 2^{62}, \quad (4.11)$$

because $h_{n_1} + h_{n_2} + h_n \leq H(2^{1-b} + 1)n^b < 4Hn^b$.

Fix $k \geq 1$. Iterating Equation (4.11) on equal-size blocks gives

$$\frac{f_{2^j k}}{2^j k} \geq \frac{f_k}{k} - C_2 k^{b-1} \sum_{i=1}^j 2^{-(1-b)i}.$$

Let

$$D_b = \sum_{i \geq 1} 2^{-(1-b)i} = \frac{1}{2^{1/6} - 1} < 32.$$

The last inequality follows from $(33/32)^6 < 2$. Thus every dyadic multiple $s = 2^j k$ satisfies $f_s \geq s(f_k/k - C_2 D_b k^{-1/6})$. For $n \geq k$, use the binary expansion of $\lfloor n/k \rfloor$ to partition $k \lfloor n/k \rfloor$ into distinct dyadic multiples of k . Merge them in increasing order of size. When a block of size $2^j k$ is added, the total merged size is less than $2^{j+1} k$; the sum of merging errors in Equation (4.11) is bounded by

$$C_2 \sum_{j=0}^{\lfloor \log_2(n/k) \rfloor} (2^{j+1} k)^b \leq A_b C_2 n^b$$

for a fixed finite constant A_b . Add the remainder, if nonzero, in one more application of Equation (4.11), using $f_s \geq 0$ for its contribution. Divide by n and let n tend to infinity with k fixed. Since $b < 1$, the merging errors vanish and

$$\liminf_{n \rightarrow \infty} \frac{f_n}{n} \geq \frac{f_k}{k} - C_2 D_b k^{-1/6} \geq \frac{f_k}{k} - 2^{67} k^{-1/6}.$$

The bounded sequence $f_n/n \in [0, B]$ has a limsup subsequence. Let k tend to infinity along that subsequence in the last inequality. Its liminf is at least its limsup, proving convergence and Equation (4.10). $\square$

4.3 The threshold in the proper-clause model

Theorem 4.4 (Limiting satisfiability threshold). *The limit α in Proposition 4.3 lies in $[1/200, 10]$. For every fixed real $a \geq 0$, the relaxed Poisson model satisfies*

$$P_n(an) \longrightarrow \begin{cases} 1, & 0 \leq a < \alpha, \\ 0, & a > \alpha. \end{cases}$$

The same limits hold for $p(n, \lfloor an \rfloor)$ in the proper-clause model defined in the introduction. In particular its limiting density is $\alpha_3 = \alpha$.

Proof. For a fixed assignment, a relaxed clause is violated with probability $1/8$, even if some variable indices repeat, because the signs are independent. The expected number of satisfying assignments at time $10n$ is therefore

$$2^n \exp(-10n/8) \longrightarrow 0.$$

Consequently $P_n(10n) \rightarrow 0$, and $f_n \leq 10n + (B - 10)nP_n(10n)$ shows that $\alpha \leq 10 < B$.

For a lower bound set $c = 1/100$ and $M = \lfloor cn \rfloor$. A union bound shows that the probability some ℓ of the first M clauses touch fewer than ℓ distinct variables is at most

$$\sum_{\ell=1}^M \binom{M}{\ell} \binom{n}{\ell} (\ell/n)^{3\ell} \leq \sum_{\ell=1}^M (e^2 c \ell/n)^\ell = o(1). \quad (4.12)$$

To obtain the first inequality, pad any set of fewer than ℓ touched variables to one of size ℓ ; all 3ℓ index choices must lie in that set. The second uses $\binom{n}{\ell} \leq (en/\ell)^\ell$. For the last limit, the terms with $\ell \leq \sqrt{n}$ sum to $O(n^{-1/2})$, and the remaining terms are at most $n(e^2 c^2)^{\sqrt{n}}$ in total. On the complementary event, Hall's marriage theorem [13, Theorem 1] assigns every clause a distinct representative variable appearing in it. Set each representative to satisfy one of its occurrences in its assigned clause. These requirements are consistent because the representatives are distinct, and they satisfy all clauses. This argument permits repeated indices and signs within a clause. At time $cn/2$ the Poisson clause count is at most M with probability tending to one. Thus $P_n(cn/2) \rightarrow 1$. Since $f_n \geq (cn/2)P_n(cn/2)$, we obtain $\alpha \geq c/2 = 1/200$.

If $a < \alpha$, convergence of f_n/n and Equation (4.3) imply $P_n(an) \rightarrow 1$. Explicitly, for $a > 0$ and sufficiently large n , $f_n - an \geq (\alpha - a)n/2$, so the failure probability is at most $C_1 n / ((\alpha - a)n/2)^p$, which tends to zero. The case $a = 0$ is immediate. If $\alpha < a < B$, the analogous upper-tail estimate gives $P_n(an) \rightarrow 0$. For $a \geq B$, use monotonicity and any intermediate density in (α, B) .

We now take $n \geq 3$ and return to distinct-variable clauses. Write $P_n^d(t)$ for their rate-one Poisson satisfiability probability. Thinning out the relaxed clauses having repeated variable indices leaves an independent distinct-clause stream of rate

$$s_n = (1 - 1/n)(1 - 2/n).$$

Removing clauses can only help satisfiability. Conversely, the discarded stream is empty up to time an with probability $\exp(-an(1 - s_n))$, independently of the retained stream. Thus

$$e^{-an(1-s_n)} P_n^d(s_n an) \leq P_n(an) \leq P_n^d(s_n an). \quad (4.13)$$

For fixed a , the exponential factor is at least e^{-3a} . If $a < \alpha$, choose $a' \in (a, \alpha)$. For large n , $s_n a' > a$, and monotonicity with Equation (4.13) gives $P_n^d(an) \geq P_n^d(a'n) \rightarrow 1$. If $a > \alpha$, choose $a' \in (\alpha, a)$. Then

$$P_n^d(an) \leq P_n^d(s_n a'n) \leq e^{3a'} P_n(a'n) \rightarrow 0.$$

Finally let $p(n, m)$ denote the probability for exactly m independently sampled distinct-variable clauses, allowing repeated clauses. It is nonincreasing in m . Below the threshold, compare $m = \lfloor an \rfloor$ with a Poisson count of mean $a'n$ for $a < a' < \alpha$. This count is at least m with probability tending to one, so $p(n, m) \geq P_n^d(a'n) - o(1) \rightarrow 1$. Above the threshold use $\alpha < a' < a$; the Poisson count is at most m with probability tending to one, giving $p(n, m) \leq P_n^d(a'n) + o(1) \rightarrow 0$. Conditioning three ordered uniform indices to be distinct and then forgetting their order produces exactly a uniform three-element subset with independent fair signs. Clauses remain independent and may repeat. This completes the transfer to the exact model. $\square$

4.4 Computing the lower certificates

The preceding theorem identifies the limit with the proper-clause threshold. We now make the one-sided bound in Equation (4.10) into finite rational certificates.

Proposition 4.5 (Effective finite-size integration). *The real f_n is computable uniformly in the integer n . There is an effective enumeration of rational numbers strictly below α whose supremum is α . In particular, there is a computable rational sequence $(\ell_n)_{n \geq 1}$ with $\ell_n < \alpha$ and $\ell_n \rightarrow \alpha$.*

Proof. For integers $n \geq 1$ and $m \geq 0$, let $s_{n,m}$ be the relaxed-model satisfiability probability with exactly m clauses. This is rational and can be computed by enumerating all $(8n^3)^m$ ordered signed clause lists and all 2^n assignments. The survival-integral identity and conditioning on the Poisson count give

$$f_n = \int_0^{Bn} P_n(t) dt = \sum_{m=0}^{\infty} s_{n,m} \int_0^{Bn} e^{-t} \frac{t^m}{m!} dt.$$

The summands are nonnegative. Uniformly for $0 \leq t \leq Bn$,

$$\mathbb{P}(\text{Poi}(t) > M) \leq e^{Bn} 2^{-(M+1)}.$$

Thus truncation at M changes the integral by at most $Bne^{Bn} 2^{-(M+1)}$, a bound that can be made smaller than any prescribed positive rational error by a finite search using the integer upper bound $e^{Bn} < 3^{Bn}$. Each remaining integral equals

$$1 - e^{-Bn} \sum_{j=0}^m \frac{(Bn)^j}{j!},$$

and is computable with certified rational intervals. For completeness, if $x = Bn$ and $J \geq 2x$ is an integer, the partial sum $S_J = \sum_{j=0}^J x^j/j!$ satisfies

$$S_J \leq e^x \leq S_J + 2 \frac{x^{J+1}}{(J+1)!}.$$

Indeed, each ratio after the first omitted term is at most $1/2$. The rational upper bound for the tail tends to zero. Taking reciprocals gives certified intervals for e^{-x} , and then for each displayed integral. Together with the explicit Poisson tail, this proves uniform computability of f_n .

Rational bisection on $[0, 1]$ computes the sixth root of $1/n$, so

$$v_n = \frac{f_n}{n} - 2^{67} n^{-1/6}$$

is uniformly computable. By Proposition 4.3, $v_n \leq \alpha$, and $v_n \rightarrow \alpha$. For each pair of positive integers (n, j) , compute a rational $\hat{v}_{n,j}$ within 2^{-j} of v_n , and output

$$\ell_{n,j} = \hat{v}_{n,j} - 2^{1-j}.$$

Every output is strictly less than v_n , hence less than α . Enumerating all pairs (n, j) is effective, and the supremum of the outputs is α . To obtain a single sequence, choose effectively $j(n)$ with $3 \cdot 2^{-j(n)} \leq 1/n$ and put $\ell_n = \ell_{n,j(n)}$. Then

$$v_n - \frac{1}{n} \leq \ell_n < v_n \leq \alpha.$$

Since $v_n \rightarrow \alpha$, this sequence also tends to α . It therefore supplies the lower sequence required by Lemma 2.1. $\square$

4.5 Robust unsatisfiability and finite-temperature witnesses

The lower certificate family is now complete. For the upper search, we need a negative pressure witness above the threshold. Exact unsatisfiability alone does not supply one: the number of violated clauses could still be sublinear. The deletion moment prevents this by bounding how much extra survival time a linear deletion allowance buys.

Proposition 4.6 (Linear deletion and violation robustness). *For all integers $n \geq 1$ and $0 \leq r \leq n$,*

$$\frac{\mathbb{E}F_r}{n} \leq \frac{f_n}{n} + C_4(r/n)^{1/6}, \quad C_4 = 2^{53}. \quad (4.14)$$

For every fixed real $a > \alpha$, there exists $\epsilon > 0$ such that, with probability tending to one, the relaxed formula at time an cannot be made satisfiable by deleting $\lfloor \epsilon n \rfloor$ variables. On the same event every assignment violates more than $\lfloor \epsilon n \rfloor$ clauses.

Proof. The case $r = 0$ is equality. For $r \geq 1$, fix a realization. The survival time with at most r deletions is the maximum of the survival times over finitely many fixed deletion sets. Capping commutes with this finite maximum. Choose a maximizing set D , padding it to size exactly r ; padding cannot worsen survival. For every $v \in D$, the free deletion of v followed by at most $r - 1$ other deletions has capped survival time exactly F_r : the choice $D \setminus \{v\}$ attains it, and every competing choice uses at most r deletions in total. Hence, for $1 \leq r \leq n$,

$$r(F_r - F_{r-1})^p \leq \sum_{v=1}^n (F_{r-1}^{+v} - F_{r-1})^p.$$

This includes the case of survival up to the cap and the case $r = n$. Take expectations and use Lemma 4.1, followed by the L^p bound on the first moment. Summing over the deletion allowance,

$$\mathbb{E}F_r - f_n \leq (Cn)^{1/p} \sum_{j=1}^r j^{-1/p} \leq 6C^{5/6} n^{5/6} r^{1/6} < 2^{53} n^{5/6} r^{1/6},$$

which proves Equation (4.14).

Fix $a > \alpha$ and choose $a_0 \in (\alpha, \min(a, B))$. Let $\Delta = a_0 - \alpha > 0$ and choose $0 < \epsilon < 1$ so small that $C_4 \epsilon^{1/6} < \Delta/4$. For large n , convergence of f_n/n and Equation (4.14), with $r = \lfloor \epsilon n \rfloor$, give $\mathbb{E}F_r \leq (\alpha + \Delta/2)n$. Since $a_0 < B$, survival at time $a_0 n$ implies $F_r > a_0 n$. Equation (4.3) therefore shows

$$\mathbb{P}(T_r > a_0 n) \leq \frac{C_1 n}{(\Delta n/2)^p} \rightarrow 0.$$

By monotonicity the same conclusion holds at time an . If an assignment violated at most r clauses, selecting one variable from each violated clause and deleting those variables would discard every violation. The remaining clauses would be satisfied by the restricted assignment, using at most r deletions. This contradiction proves the violation assertion. $\square$

For a relaxed formula at time an , let $H_n(\sigma)$ be the number of clauses violated by σ , and define

$$Z_n(a, \beta) = \sum_{\sigma \in \{0,1\}^n} e^{-\beta H_n(\sigma)}, \quad \beta > 0.$$

Corollary 4.7 (Signs of the soft pressure). *For every fixed real $0 \leq a < \alpha$ and every fixed $\beta > 0$,*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \log Z_n(a, \beta) \geq 0. \quad (4.15)$$

For every fixed real $a > \alpha$, some positive integer β satisfies

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \log Z_n(a, \beta) < 0. \quad (4.16)$$

These statements concern the finite-size expected logarithms and do not need existence of a pressure limit. Once that limit is proved, they give the two signs used in the upper certificate search.

Proof. Put $V_n = \min_{\sigma} H_n(\sigma)$, and let $M_n \sim \text{Poi}(an)$ be the total clause count. Below the threshold, $\mathbb{P}(V_n > 0) \rightarrow 0$ and $0 \leq V_n \leq M_n$. Cauchy–Schwarz gives

$$\frac{\mathbb{E}V_n}{n} \leq ((a^2 + a/n)\mathbb{P}(V_n > 0))^{1/2} \rightarrow 0.$$

The bound $\log Z_n \geq -\beta V_n$ proves Equation (4.15). Above the threshold, Proposition 4.6 gives $\liminf_n \mathbb{E}V_n/n \geq \epsilon$ for some $\epsilon > 0$. Since $\log Z_n \leq n \log 2 - \beta V_n$, any integer $\beta > (\log 2)/\epsilon$ proves Equation (4.16). $\square$

5 Finite trials for the pressure

The upper branch of the certificate search needs finite descriptions whose values can be evaluated and which bound the pressure from above. We define those descriptions first. A local three-literal interpolation will prove their upper bound. Finite formulas will then supply nearly optimal trials, although initially with an extra source of randomness shared by the sites. The final part of this section gives the exact change of law used later to remove that dependence.

Fix a density $a > 0$ and a finite penalty $\beta > 0$. In this section the formula has $\text{Poi}(an)$ clauses, and the three variable positions of each clause are independent and uniform in $[n]$, with independent fair signs. Thus positions may repeat. If $H_n(\sigma)$ is the number of clauses violated by $\sigma \in \{0,1\}^n$, put

$$Z_n = \sum_{\sigma \in \{0,1\}^n} e^{-\beta H_n(\sigma)}, \quad L_n = \mathbb{E} \log Z_n, \quad w = 1 - e^{-\beta}. \quad (5.1)$$

We will prove that L_n/n has a finite limit $P(a, \beta)$. The parameter w lies in $(0, 1)$: a violated clause multiplies the weight by $1 - w = e^{-\beta}$.

5.1 Trial descriptions and their evaluation

A trial consists of a finite hierarchy of conditional averages and a rule that produces a pair of numbers in $[0, 1]$ at each site. The site index distinguishes copies of the sampling generators defined below. The depth is an integer $d \geq 1$, and its averaging exponents satisfy

$$0 < t_1 \leq \dots \leq t_d \leq 1. \quad (5.2)$$

Thus intermediate trials may have equal exponents or exponent one. A trial is called *strict* when $0 < t_1 < \dots < t_d < 1$.

The sampling data are as follows. A root variable z_0 has an arbitrary law on a standard Borel space; it stores randomness common to all sites. At each level $j \in \{1, \dots, d\}$, a uniform variable $z_j \in [0, 1]$ is also shared by all sites. A site v has uniform variables $u_0^v, \dots, u_d^v \in [0, 1]$. All these uniform generators are mutually independent and independent of z_0 . The same two measurable functions at every site give the messages

$$f^\pm(z_0, z_1, \dots, z_d; u_0^v, u_1^v, \dots, u_d^v) \in [0, 1]. \quad (5.3)$$

The two coordinates are indexed by the sign of a literal; they need not sum to one. General conditional sampling kernels can be encoded by these uniform generators, so the resulting messages need not be independent. Each use of a message pair in a clause will be called a reservoir occurrence. It receives a fresh site, including when another occurrence is in the same clause. Distinct fresh generators may of course produce equal sampled values.

For $t > 0$ and a bounded random variable X , write

$$T_t X = \frac{1}{t} \log \mathbb{E} e^{tX}$$

when the variables integrated by the expectation have been specified. For a loss depending on finitely many sites, define

$$\mathcal{T}X = T_{t_1} \dots T_{t_d} X. \quad (5.4)$$

The rightmost operation acts first. At level j , it integrates z_j and all the sites' u_j^v , conditional on the preceding levels. The root variable, all u_0^v , and any further root data are held fixed. Conditional expectations make this rule meaningful even when some generators are unused. For the losses below, the root clause counts give a bound on X , so the same definition applies after conditioning on those counts.

An A-packet has a count $D \sim \text{Poi}(3a)$. Each of its D clauses contains one common real Boolean variable b , two fresh reservoir occurrences, and three independent fair signs. Let $I_h(b)$ indicate that the real literal of clause h is false at b , and let f_{h1}, f_{h2} be the messages chosen by the two reservoir signs. Its loss is

$$X_A = \log \sum_{b=0,1} \prod_{h=1}^D (1 - w I_h(b) f_{h1} f_{h2}). \quad (5.5)$$

A B-packet has a count $D \sim \text{Poi}(2a)$ and three fresh reservoir occurrences with independent fair signs in every clause. Its loss is

$$X_B = \sum_{h=1}^D \log(1 - w f_{h1} f_{h2} f_{h3}). \quad (5.6)$$

Counts and signs are root data, independent of the trial generators and z_0 ; different packets use independent such data. Each clause factor in Equations (5.5)–(5.6) lies in $[e^{-\beta}, 1]$. In particular,

$$\log 2 - \beta D \leq X_A \leq \log 2, \quad -\beta D \leq X_B \leq 0, \quad |X_A|, |X_B| \leq \log 2 + \beta D. \quad (5.7)$$

The conventions for $D = 0$ give $X_A = \log 2$ and $X_B = 0$.

For a positive integer k , use fresh sites and independent clause data in each of k packets and define the *total* value

$$G_k = \mathbb{E} \mathcal{T} \sum_{i=1}^k X_{A,i} - \mathbb{E} \mathcal{T} \sum_{i=1}^k X_{B,i}. \quad (5.8)$$

The outer expectations integrate all root data. A trial is *site-only* if its message functions do not use $z_1, \dots, z_d$. The shared root variable z_0 remains allowed. For such a trial, conditional on the root data, the site variables of distinct packets are independent at each level. The exponential of a sum therefore factors at every conditional average, and subsequent root expectation gives

$$G_k = kG_1 \quad \text{for every site-only trial.} \quad (5.9)$$

For a general trial, G_k remains a total value; the shared nonroot variables may couple the packets, and we do not use this additivity.

Here is the finite class used by the algorithm. A *rational trial* is a strict site-only trial in which z_0 is uniform on $[0, 1]$, all exponents are rational, and the message pair is a rational-valued step function on a finite rational rectangular grid in

$$(z_0, u_0, u_1, \dots, u_d) \in [0, 1]^{d+2}.$$

More explicitly, choose a finite increasing list of rational endpoints from 0 to 1 in each coordinate and give a rational pair in $[0, 1]^2$ for every resulting rectangle. Intervals are half open except at the right endpoint 1. The common coordinate z_0 is sampled once and is shared by every site. Let $\mathcal{Q}$ denote this class. Its finite tables, grids, and strictly ordered exponent lists have an effective enumeration by finite strings; their validity is checked by rational comparisons.

Lemma 5.1 (Evaluation of a fixed rational trial). *For rational $a > 0$, a positive integer β , and a finite description $Q \in \mathcal{Q}$, the value $G_1(a, \beta; Q)$ is uniformly computable from these data. In particular, strict negativity of this value is semidecidable by certified rational upper bounds.*

Proof. Fix the count and signs of one packet. Every reservoir occurrence uses its own site coordinates; only the root coordinate is common. Subdividing the finitely many grids expresses each conditional integral as a finite sum with rational weights. The nested transforms and the remaining root average are consequently finite expressions using rational arithmetic, exponentials, and logarithms. Every exponent is positive, and every clause factor is at least $e^{-\beta}$. Thus all logarithms have strictly positive arguments, and these operations admit uniform rational interval approximations to any prescribed accuracy. The finite sign average and the Poisson probabilities are computable in the same way.

It remains to control the count tail uniformly over the description. The conditional transforms preserve the bounds in Equation (5.7). For $D \sim \text{Poi}(\lambda)$ and any integer $M \geq 0$,

$$\begin{aligned} \mathbb{E}[(\log 2 + \beta D) \mathbf{1}_{\{D > M\}}] &\leq 2^{-M} \mathbb{E}[(\log 2 + \beta D) 2^D] \\ &= 2^{-M} e^{\lambda(\log 2 + 2\beta\lambda)}, \end{aligned} \quad (5.10)$$

where $\lambda = 3a$ for the A-packet and $\lambda = 2a$ for the B-packet. Choose M so that the sum of these two computable bounds is smaller than the desired tail error, and then approximate the finitely many retained terms. This gives a rational interval containing G_1 with arbitrarily small width. Its upper endpoint eventually becomes negative exactly when $G_1 < 0$. $\square$

5.2 A local interpolation bound

We first establish the pressure limit in the same relaxed clause model. The proof uses only that a fresh clause chooses its three signed positions independently. It also provides the logarithmic expansion used for trial upper bounds.

Lemma 5.2 (Pressure limit). *For every $a > 0$ and finite $\beta > 0$, the sequence L_n in Equation (5.1) is superadditive. Hence the finite limit*

$$P(a, \beta) = \lim_{n \rightarrow \infty} \frac{L_n}{n} = \sup_{n \geq 1} \frac{L_n}{n} \quad (5.11)$$

exists and satisfies $\log 2 - \beta a \leq P(a, \beta) \leq \log 2$.

Proof. Partition $n = n_1 + n_2$ variables into two nonempty blocks and put $\lambda_i = n_i/n$. At $s \in [0, 1]$, take independent within-block clauses of means $(1-s)an_i$ and global clauses of mean san , and write Z_s for the resulting partition function. Brackets denote the current Gibbs distribution, with independent replicas when several samples occur. Adding a clause with violation indicator I changes the log partition function by

$$\log(1 - w\langle I \rangle) = - \sum_{\ell \geq 1} \frac{w^\ell}{\ell} \langle I \rangle^\ell. \quad (5.12)$$

The series is uniformly absolutely summable because $0 < w < 1$.

For fixed replicas $\sigma^1, \dots, \sigma^\ell$, let x_i be the fraction of the $2n_i$ signed literals in block i that are false in every replica. Then $x_i \in [0, 1]$. A fresh within-block clause has mean replica product x_i^3 , whereas a fresh global clause has mean replica product $(\lambda_1 x_1 + \lambda_2 x_2)^3$. Poisson differentiation and Equation (5.12) therefore give

$$\frac{d}{ds} \mathbb{E} \log Z_s = an \sum_{\ell \geq 1} \frac{w^\ell}{\ell} \mathbb{E} \left\langle \lambda_1 x_1^3 + \lambda_2 x_2^3 - (\lambda_1 x_1 + \lambda_2 x_2)^3 \right\rangle \geq 0.$$

The last inequality is convexity of the cube on $[0, 1]$. Differentiation is justified because one added clause changes $\log Z_s$ by at most β , and the series has the uniform bound just noted. At $s = 0$ the partition function factors into the two block partition functions; at $s = 1$ it has the size- n law. Thus $L_n \geq L_{n_1} + L_{n_2}$.

If the formula has M clauses, then $n \log 2 - \beta M \leq \log Z_n \leq n \log 2$. Taking expectations gives $n(\log 2 - \beta a) \leq L_n \leq n \log 2$. The superadditive lemma now proves Equation (5.11). The argument specializes sparse interpolation from [3]; its nonnegative signed-literal proportions give the cubic convexity used here. $\square$

Proposition 5.3 (Sound trial upper bounds). *For every $a > 0$ and finite $\beta > 0$, each finite-depth strict site-only trial satisfies*

$$L_n \leq nG_1 \quad (n \geq 1), \quad P(a, \beta) \leq G_1. \quad (5.13)$$

We prove this by expressing the nested averages as averages over random weighted paths. These weights form a Ruelle probability cascade (RPC), originating in [29]. The marked Poisson change of measure below is the cascade reweighting mechanism of Panchenko and Talagrand [28, Lemmas 2.1 and 3.1]. We give the derivation because both its conditional marks and its partition prediction rule will be used again in the structural argument.

Lemma 5.4 (Cascade identities). *For $0 < t_1 < \dots < t_d < 1$, there are random positive weights $(v_\alpha)_{\alpha \in \mathbb{N}^d}$ with $\sum_\alpha v_\alpha = 1$ and the following properties.*

Independent samples from these weights give nested exchangeable partitions by their common prefixes. Conditional on the partitions of $n \geq 1$ sampled leaves, the probability that the next sample belongs to a specified already seen level- j cluster C , for $1 \leq j \leq d$, is

$$\frac{n_C - t_j}{n}, \quad (5.14)$$

where n_C is the number of samples in C . These probabilities, and their differences between parent and child clusters, determine the partition law. The ranked cluster masses are the limiting sample frequencies.

Take any root data independent of the cascade weights. Conditional on those data, attach independent uniform marks to the nonroot vertices, independently of the weights; any root marks are included in the root data. Each mark may be a vector of uniforms. Let Y_α be the value of the same bounded measurable function of the path marks and root data. Then

$$\mathbb{E} \log \sum_{\alpha \in \mathbb{N}^d} v_\alpha e^{Y_\alpha} = \mathbb{E} T_{t_1} \dots T_{t_d} Y_{\text{path}}. \quad (5.15)$$

This identity holds conditionally on the root data. It also holds when $|Y_\alpha| \leq B$ for all leaves, where B depends only on the root data, is finite almost surely, and satisfies $\mathbb{E}B < \infty$.

Proof. At every vertex of depth $j - 1$, independently place a Poisson process of positive child points with intensity $t_j u^{-t_j-1} du$. The unnormalized weight W_α of a leaf is the product of the points along its path. To verify that their total mass is finite, suppose a child subtree has total mass S' measured from that child. For $x = t_j$, the mapping $u \mapsto v = uS'$ turns its child process into one of intensity

$$cxv^{-x-1} dv, \quad c = \mathbb{E}(S')^x.$$

The attached subtree law is tilted by $(S')^x/c$ and is independent of the mapped point. This is the marked Poisson mapping formula. At a leaf, $S' = 1$.

If S is the sum of a process with intensity $cxv^{-x-1} dv$, then

$$\mathbb{E} e^{-qS} = \exp\{-c\Gamma(1-x)q^x\}, \quad q > 0. \quad (5.16)$$

It follows that $0 < S < \infty$ almost surely. For $s \geq 1$,

$$\mathbb{P}(S > s) \leq \frac{1 - \mathbb{E} e^{-S/s}}{1 - e^{-1}} \leq C_{c,x} s^{-x},$$

with a finite constant $C_{c,x}$. For $0 < s \leq 1$,

$$\mathbb{P}(S \leq s) \leq e \mathbb{E} e^{-S/s} = e \exp\{-c\Gamma(1-x)s^{-x}\}.$$

Consequently $\mathbb{E} S^y < \infty$ for $0 \leq y < x$, and $\mathbb{E} |\log S| < \infty$. Since the exponents strictly increase down the tree, induction from the leaves gives the required moment $\mathbb{E}(S')^{t_j} < \infty$ at every vertex. The total leaf mass is therefore finite and positive; normalize it to obtain $v_\alpha = W_\alpha / \sum_\gamma W_\gamma$.

We next derive the prediction rule, including the effect of selection through a parent. Consider a node whose child exponent is x and whose parent exponent is $y < x$, with $y = 0$ at the root. Once this node is selected from its parent, its point-sum law is tilted by $S^y / \mathbb{E} S^y$. Its subtree marks remain independent of the mapped child points, with the individually tilted laws described above.

Write (V_i) for the mapped points, so that $S = \sum_i V_i$. For a specified partition of n samples into h nonempty children with sizes $n_1, \dots, n_h$, the probability is

$$\begin{aligned} & \frac{1}{\mathbb{E}S^y} \mathbb{E} \sum_{i_1, \dots, i_h \text{ distinct}} S^{y-n} \prod_{j=1}^h V_{i_j}^{n_j} \\ &= \frac{\prod_{j=1}^h (cx\Gamma(n_j - x))}{\mathbb{E}S^y \Gamma(n - y)} \int_0^\infty q^{hx-y-1} e^{-c\Gamma(1-x)q^x} dq. \end{aligned} \quad (5.17)$$

Indeed, the Gamma integral represents $S^{-(n-y)}$, and the Poisson factorial-moment formula evaluates the sum over distinct points. Increasing n_j by one changes the displayed expression by the factor $(n_j - x)/(n - y)$. The complementary probability of a new child is $(hx - y)/(n - y)$. Observations deeper in the tree concern independent subtree marks and do not change these point predictions. Along an already seen chain, multiplication of the within-parent probabilities telescopes to Equation (5.14). In the usual Poisson–Dirichlet notation, the child partition seen through its parent has law $\text{PD}(x, -y)$, rather than the fresh $\text{PD}(x, 0)$ law. The finite prediction probabilities determine the nested partition laws, and conditional independent sampling with the strong law recovers their cluster masses as frequencies.

For Equation (5.15), condition on root data and put $y_d = Y$ and $y_{j-1} = T_{t_j} y_j$ recursively, integrating the level- j marks. At the last level, map a Poisson point u to ue^{y_d} . Conditional on the ancestor marks, the mapped process is the original process scaled by e^{y_d-1} , since its intensity is multiplied by $\mathbb{E}e^{t_d y_d} = e^{t_d y_d-1}$. Discard the integrated marks. After the scale is divided out, the remaining descendant total has its unmarked law and is independent of the retained ancestor marks. Repeating this mapping at the preceding levels gives, conditionally on the root data,

$$\sum_{\alpha} W_{\alpha} e^{Y_{\alpha}} \stackrel{\text{law}}{=} e^{T_{t_1} \dots T_{t_d} Y_{\text{path}}} \sum_{\alpha} W_{\alpha}.$$

This is an equality of marginal distributions. Taking mean logarithms and subtracting the mean logarithm of the unmarked total proves Equation (5.15); log integrability was established above. For a root-conditionally bounded Y , apply the identity at each admissible root value and then integrate its bound.

Children may initially be ordered by raw Poisson points or by total subtree masses. These permutations depend only on the unmarked cascade and preserve ancestry, so independent hierarchical marks have the same conditional law in either ordering. A permutation made after multiplication by e^Y need not preserve the law of retained marks. The backward calculation discards exactly the marks it integrates and makes no such independence assertion. $\square$

The comparison uses the diluted interpolation method of Franz and Leone and of Panchenko and Talagrand [8, 27], which extends Guerra’s interpolation for Gaussian spin glasses [11, arXiv version, Theorems 3 and 5]. Panchenko and Talagrand’s general finite-step theorem is stated for even arity. For this even-arity class, the companion [20, Theorem 2.1 and Corollary 10.2] proves the exact finite-hierarchy variational formula under the stated Panchenko–Talagrand hypotheses, including soft even- K SAT in the Poisson model with independently sampled variable positions. Its cavity functional is separate from the three-literal trial functional used here. For K-SAT, Talagrand observed that the nonnegative literal factors make the convexity step valid at every clause size [31, author manuscript, Section 6.5, Equations (6.120) and (6.129)]. We verify the three-literal calculation in the present trial model, including the independence required for distinct reservoir occurrences.

Proof of Proposition 5.3. Use the cascade of Lemma 5.4. For each reservoir occurrence, attach a fresh tree of site marks and a fresh root site uniform. All these trees are independent conditional on the root global z_0 ; the site-only message functions use no nonroot global marks. At $s \in [0, 1]$, take independent Poisson collections of three kinds of factors:

kind	mean count	factor
three real positions	$(1-s)an$	$1 - wI$
three reservoir occurrences	$(1-s)2an$	$1 - wf_1f_2f_3$
one real position, two reservoir occurrences	$s3an$	$1 - wI_0f_1f_2$.

Real indices are uniform in $[n]$, with independent positions in the first kind; all signs are independent and fair. Here I is the violation indicator of a real clause and I_0 indicates that the one real literal is false. Let $\mathcal{Z}_s$ be the sum over real assignments σ and cascade leaves α of v_α times the product of these factors, using the marks on path α for its messages.

A Gibbs replica for this partition sum consists of a real assignment and a cascade label. Conditional on ℓ replicas and the existing data, including z_0 , let $x \in [0, 1]$ be the fraction of real signed literals false in all their assignments. Let $y \in [0, 1]$ be the expected product of the ℓ signed messages at one fresh reservoir occurrence evaluated at their labels. This expectation includes its fresh root uniform, its fresh tree of site marks, and one fresh fair sign; within the occurrence, the replicas use the marks on their respective paths. Distinct occurrences use independent copies of these data. Hence the mean replica products for the three rows of the table are respectively x^3 , y^3 , and xy^2 .

Poisson differentiation and Equation (5.12) now give

$$\frac{d}{ds} \mathbb{E} \log \mathcal{Z}_s = an \sum_{\ell \geq 1} \frac{w^\ell}{\ell} \mathbb{E} \langle x^3 + 2y^3 - 3xy^2 \rangle \geq 0,$$

because

$$x^3 + 2y^3 - 3xy^2 = (x - y)^2(x + 2y) \geq 0. \quad (5.18)$$

Each new factor lies in $[e^{-\beta}, 1]$, so the same bounded-increment and absolute-convergence arguments justify the differentiation. The fresh-site factorization just used is the one supplied by the site-only class. We apply this upper-bound argument only to that class, while later intermediate trials retain their used nonroot shared variables.

At $s = 0$, the real partition sum separates from the cascade sum. Splitting the reservoir clause process into n independent B-packets gives

$$\mathbb{E} \log \mathcal{Z}_0 = L_n + \mathbb{E} \mathcal{T} \sum_{i=1}^n X_{B,i}$$

by Equation (5.15). At $s = 1$, splitting by the uniform real index gives an independent A-packet at each index. The sum over real bits then factors before the cascade sum, and the same identity gives

$$\mathbb{E} \log \mathcal{Z}_1 = \mathbb{E} \mathcal{T} \sum_{i=1}^n X_{A,i}.$$

The nonnegative derivative therefore proves $L_n \leq G_n = nG_1$, where the last equality is Equation (5.9). Taking the pressure limit proves the second inequality in Equation (5.13). $\square$

5.3 Finite reservoirs give nearly optimal values

We next obtain trials from finite formulas. The comparison with a finite reservoir parallels the cavity increment in the variational principle of Aizenman, Sims, and Starr for a Gaussian class including the Sherrington–Kirkpatrick model [1, arXiv version, Lemma 3]. We compute the Poisson clause rates directly. The construction gives a value per packet at most $P + \epsilon$ for any chosen $\epsilon > 0$, but uses one nonroot variable shared by all sites. That variable will have to be removed before the trial belongs to $\mathcal{Q}$.

Lemma 5.5 (Finite-reservoir approximation). *Fix $a > 0$ and finite $\beta > 0$. For each positive integer k and each $\epsilon > 0$, there is a depth-one trial with $t_1 = 1$ such that*

$$\frac{G_k}{k} \leq P(a, \beta) + \epsilon. \quad (5.19)$$

Consequently there is a sequence of trials, one for each horizon $N \rightarrow \infty$, with $G_N/N \leq P(a, \beta) + o(1)$. These trials may use a shared nonroot global variable, and the trial at one horizon need not be the trial at another.

Proof. Fix positive integers M, k and split a size- $M + k$ formula into M old and k new variables. Its all-old clauses have mean

$$\lambda_{M,k} = a(M + k) \left(\frac{M}{M + k} \right)^3 = \frac{aM^3}{(M + k)^2}. \quad (5.20)$$

Use this old formula as the root disorder z_0 . A uniform generator z_1 samples an assignment from its Gibbs law and is shared by all sites. A reservoir occurrence samples an independent uniform old index using its root site variable u_0 , and its two messages are the hard indicators that the corresponding signed literals are false in that assignment. Its level-one site variable may be ignored. Thus the depth-one transform T_1 is exactly a log Gibbs average over the old assignment. Each occurrence samples its own old index; accidental repetitions agree with the independent position law of the formula.

Clauses with exactly one new position have total mean $3akM^2/(M + k)^2$ and split uniformly among the new indices. Clauses with at least two new positions have mean

$$\mu_{\geq 2} = \frac{ak^2(3M + k)}{(M + k)^2}.$$

If the latter clauses are omitted, summing over the new bits gives the log Gibbs average of a product of k one-bit sums. Replacing the exactly-one-new mean by $3a$ at each new index produces the k A-packets. The excess mean in this replacement is

$$\Delta_A = 3ak - \frac{3akM^2}{(M + k)^2} = \frac{3ak^2(2M + k)}{(M + k)^2}.$$

On the other hand, a size- M formula is obtained from the same old formula by adding old clauses of mean $aM - \lambda_{M,k}$. Replacing this mean by $2ak$ produces the k B-packets, with excess

$$\Delta_B = 2ak - (aM - \lambda_{M,k}) = \frac{ak^2(3M + 2k)}{(M + k)^2}.$$

All these replacements can be coupled by adding independent Poisson clauses. An added factor lies in $[e^{-\beta}, 1]$, so it changes the log of any of the relevant positive partition sums by at most β . At

the ideal rates, the two mean log Gibbs increments from the same old formula are exactly the A and B terms of G_k for this trial, which depends on (M, k) . Comparing the original and ideal rates therefore gives

$$\begin{aligned} |L_{M+k} - L_M - G_k| &\leq \beta(\mu_{\geq 2} + \Delta_A + \Delta_B) \\ &= \frac{a\beta k^2(12M + 6k)}{(M + k)^2} \leq \frac{12a\beta k^2}{M}. \end{aligned} \quad (5.21)$$

For fixed k , the pressure limit implies

$$\liminf_{M \rightarrow \infty} (L_{M+k} - L_M) \leq kP(a, \beta).$$

Indeed, if all sufficiently large increments exceeded kP by one fixed positive amount, summing along any arithmetic progression of step k would contradict $L_n/n \rightarrow P$. Choose M large along this liminf so that the increment divided by k is at most $P + \epsilon/2$ and the last term of Equation (5.21), divided by k , is at most $\epsilon/2$. This proves Equation (5.19). Finally, apply the result separately at $k = N$ with any error sequence tending to zero. The reservoir size may depend on N and on that error; no uniform choice is needed. $\square$

The reservoir trial uses the shared level-one Gibbs assignment and has $t_1 = 1$, so it is not yet a rational trial. The remaining approximation argument replaces its used nonroot shared variables by independent site sampling, makes the exponents strict, and then approximates the messages by finite rational tables. Theorem 8.4 will establish

$$P(a, \beta) = \inf_{Q \in \mathcal{Q}} G_1(a, \beta; Q).$$

Together with Proposition 5.3, this also gives the same infimum over all finite-depth strict measurable site-only trials. The distinction between soundness of each allowed trial and completeness of the class is essential to the certificate search.

5.4 Exact changes of path law

The next identity compares two packet collections while leaving unused site generators fresh. Its recursive densities are the cascade changes of law studied in [28, Section 3]. We prove the identity directly for all exponents in Equation (5.2), including coincident exponents and the value one.

Lemma 5.6 (Path posterior). *Let Y use a collection of old sites and let X use fresh sites in one trial, with both losses bounded conditional on the root data. Define*

$$Y_\ell = T_{t_{\ell+1}} \cdots T_{t_d} Y \quad (0 \leq \ell \leq d), \quad Y_d = Y, \quad Y_0 = \mathcal{T}Y.$$

The difference $\mathcal{T}(Y + X) - \mathcal{T}Y$ is the nested transform of X when the successive conditional laws at level ℓ are changed by the density

$$\exp\{t_\ell(Y_\ell - Y_{\ell-1})\}. \quad (5.22)$$

The old site variables may be stored as additional global variables. In this representation all fresh site generators retain their independent uniform laws. Any property of the original fresh-site kernels and their conditional uniform site laws that holds for almost every complete global path survives this change of law.

Proof. Condition on the root data, and let $\mathcal{H}_\ell$ be the history through level ℓ . By the definition of $Y_{\ell-1}$, the density in Equation (5.22) integrates to one conditional on the history through level $\ell - 1$. Put

$$V_\ell = T_{t_{\ell+1}} \cdots T_{t_d}(Y + X) - Y_\ell, \quad V_d = X.$$

At each level the identity

$$V_{\ell-1} = \frac{1}{t_\ell} \log \mathbb{E} \left[e^{t_\ell(Y_\ell - Y_{\ell-1})} e^{t_\ell V_\ell} \mid \mathcal{H}_{\ell-1} \right]$$

follows by subtracting $Y_{\ell-1}$ from the two logarithms. Applying it successively outward proves the asserted nested transform.

Each density uses only globals and old-site variables, because Y does. It leaves the product uniform law of all fresh site generators unchanged. The old-site variables can be included in the enlarged global history, which is a standard Borel random object and can again be sampled by successive uniform generators. Its finite-path law is absolutely continuous with respect to the original global and old-site law. Hence any original null set of complete global paths remains null, while the fresh-site kernels and their conditional uniform laws stay the same. The argument holds for almost every root value, as required. $\square$

Write $m = t_1$. The same operation permits a term $h(z_0, z_1)/m$ at both endpoints whenever the corresponding exponential normalizer is finite and positive and the endpoint expressions are integrable. This term passes unchanged through every level after the first. First form the posterior from Y ; then tilt its first enlarged global law by e^h , normalized conditional on the root. Conditional on the preceding history, the later level laws are unchanged. Fresh site generators still have their original laws, because this additional density uses no fresh site variable.

We will use the following consequence to pass from a large horizon to one fixed packet block. Partition N packets into complete blocks of k and a remainder of size less than k . Telescope from the all-B endpoint to the all-A endpoint, changing one block at a time and including the same h/m at both endpoints when it is present. For a complete block, the remaining packets play the role of Y . Lemma 5.6 expresses the averaged increment as the G_k functional of their posterior trial. The A and B versions of the fresh block have the same base posterior and use fresh clause data and fresh sites. These data are sampled after the base is fixed; no fresh outcome is part of its conditioning.

For each conditioned base and perturbation, let Δ_k^{post} denote this averaged block increment. The packet bounds give

$$|\Delta_k^{\text{post}}| \leq kC_{a,\beta}, \quad C_{a,\beta} = \log 2 + 5a\beta. \quad (5.23)$$

Indeed, the transformed A endpoint has absolute value at most $k \log 2 + \beta \sum D_{A,i}$ and the transformed B endpoint at most $\beta \sum D_{B,i}$; the mean counts are $3ak$ and $2ak$. The remainder has the same bound with its size, hence at most $kC_{a,\beta}$. These statements concern conditional log transforms and require neither strict exponents nor a cascade representation for the intermediate trials.

Lemma 6.1 next controls changes of all averaging exponents without a factor depending on the depth. Section 7 then constructs the pathwise continuation records used by Lemma 7.1. The posterior identity above explains why such records, once established, remain valid under later changes of the old packet data.

6 Changing the averaging parameters without paying for depth

The finite reservoir construction provides nearly optimal trials with one shared sampling level. Later intermediate trials can retain shared sampling variables at several levels. The next two sections

describe and then remove that sharing. Before introducing their structural bookkeeping, we need an estimate that controls a simultaneous change of the averaging parameters. Estimating one level at a time would charge for the number of levels, which is not known when the approximation tolerances are chosen. The estimate below avoids that dependence.

Fix the root data of a finite-depth trial and work under its conditional law. Let $\mathcal{F}_0 \subseteq \dots \subseteq \mathcal{F}_d$ be its history filtration, with $\mathcal{F}_0$ trivial under this conditional law. For a bounded $\mathcal{F}_d$ -measurable variable X and parameters $0 < t_1 \leq \dots \leq t_d \leq 1$, define

$$Y_d = X, \quad Y_{l-1} = \frac{1}{t_l} \log \mathbb{E}[e^{t_l Y_l} \mid \mathcal{F}_{l-1}], \quad 1 \leq l \leq d, \quad \mathcal{T}_{\mathbf{t}} X = Y_0.$$

The conditional sampling kernels are fixed while the parameters vary. For a single unconditional operation, write $T_t V = t^{-1} \log \mathbb{E} e^{tV}$ for $t > 0$ and $T_0 V = \mathbb{E} V$.

Lemma 6.1 (Sensitivity independent of depth). *Suppose $|X| \leq C$ almost surely, where $C \geq 0$. For any two nondecreasing vectors $\mathbf{t}, \mathbf{t}' \in (0, 1]^d$,*

$$|\mathcal{T}_{\mathbf{t}} X - \mathcal{T}_{\mathbf{t}'} X| \leq K(C) \|\mathbf{t} - \mathbf{t}'\|_{\infty}, \quad K(C) = \frac{5}{2} C^2 e^{4C}. \quad (6.1)$$

For a single operation and $0 \leq t \leq 1$,

$$0 \leq T_t X - \mathbb{E} X \leq \frac{1}{2} C^2 e^{2C} t. \quad (6.2)$$

Both assertions hold conditionally on almost every root value. Their constants are independent of the depth and of the fixed sampling kernels.

Proof. The case $C = 0$ is immediate. In general every Y_l lies in $[-C, C]$. Conditional Jensen gives nonnegative drifts

$$D_l = Y_{l-1} - \mathbb{E}[Y_l \mid \mathcal{F}_{l-1}].$$

Their expectations telescope to $\sum_l \mathbb{E} D_l = Y_0 - \mathbb{E} X \leq 2C$. Since a conditional mean minimizes mean square distance to a constant,

$$\begin{aligned} \text{Var}(Y_l \mid \mathcal{F}_{l-1}) &\leq \mathbb{E}[(Y_l - Y_{l-1})^2 \mid \mathcal{F}_{l-1}] \\ &= \mathbb{E}[Y_l^2 \mid \mathcal{F}_{l-1}] - Y_{l-1}^2 + 2Y_{l-1} D_l \\ &\leq \mathbb{E}[Y_l^2 \mid \mathcal{F}_{l-1}] - Y_{l-1}^2 + 2C D_l. \end{aligned}$$

Taking expectations and summing gives the depth-independent variance budget

$$\sum_{l=1}^d \mathbb{E} \text{Var}(Y_l \mid \mathcal{F}_{l-1}) \leq \mathbb{E} X^2 - Y_0^2 + 2C(Y_0 - \mathbb{E} X) \leq 5C^2. \quad (6.3)$$

This expectation is under the original conditional root law.

We now calculate the effect of one parameter. For a fixed bounded random variable V , let $\psi(t) = \log \mathbb{E} e^{tV}$. Under exponential tilting by $e^{tV}/\mathbb{E} e^{tV}$, one has $\psi''(t) = \text{Var}_t(V)$. Hence

$$\frac{d}{dt} T_t V = \frac{t\psi'(t) - \psi(t)}{t^2} = \frac{1}{t^2} \int_0^t u \text{Var}_u(V) du.$$

If $|V| \leq C$ and $0 \leq u \leq 1$, the tilted density is at most e^{2C} . Using the untilted mean as a competitor for the tilted variance gives $\text{Var}_u(V) \leq e^{2C} \text{Var}(V)$. Therefore

$$0 \leq \frac{d}{dt} T_t V \leq \frac{1}{2} e^{2C} \text{Var}(V). \quad (6.4)$$

The same estimate holds conditionally at each history.

When t_l varies, Y_l is fixed because it uses only the deeper parameters. Each outer operation averages the resulting change against its normalized exponential density. The chain rule therefore gives

$$\frac{\partial Y_0}{\partial t_l} = \mathbb{E}[W_{l-1} g_l], \quad g_l = \frac{d}{dt} \left(\frac{1}{t} \log \mathbb{E}[e^{tY_l} \mid \mathcal{F}_{l-1}] \right) \Big|_{t=t_l},$$

where $W_0 = 1$ and

$$W_b = \exp \left(\sum_{j=1}^b t_j (Y_j - Y_{j-1}) \right).$$

These are densities relative to the original history law; the effective values Y_j need not be independent. Summation by parts gives

$$\begin{aligned} \log W_b &= t_b Y_b - t_1 Y_0 - \sum_{j=1}^{b-1} (t_{j+1} - t_j) Y_j \\ &\leq C \left(t_b + t_1 + \sum_{j=1}^{b-1} (t_{j+1} - t_j) \right) = 2C t_b \leq 2C. \end{aligned}$$

The nondecreasing order of the parameters is used in this inequality. Combining it with Equations (6.4) and (6.3) yields

$$\sum_{l=1}^d \left| \frac{\partial Y_0}{\partial t_l} \right| \leq \frac{1}{2} e^{4C} \sum_{l=1}^d \mathbb{E} \operatorname{Var}(Y_l \mid \mathcal{F}_{l-1}) \leq \frac{5}{2} C^2 e^{4C}.$$

The straight segment joining $\mathbf{t}$ and $\mathbf{t}'$ consists of nondecreasing vectors in $(0, 1]^d$. Integrating the directional derivative proves Equation (6.1). Equal adjacent parameters cause no problem: the nested transforms are differentiable in every positive coordinate, and the density estimate holds along the entire segment.

For one operation, Equation (6.4) and $\operatorname{Var}(X) \leq C^2$ give the upper bound in Equation (6.2) after integration from 0 to t . Jensen gives the lower bound, and bounded convergence gives $T_t X \rightarrow \mathbb{E} X$ as $t \downarrow 0$. Boundedness justifies all conditional differentiations. The argument holds under each conditional root law for which the kernels are defined, hence for almost every root value. $\square$

Two consequences will be used after the structural part. First, consider pairwise disjoint consecutive blocks

$$I_j = \{b_j + 1, \dots, c_j\} \subseteq \{1, \dots, d\}, \quad t_{c_j} - t_{b_j+1} \leq \eta.$$

Replace all parameters in I_j by t_{c_j} . The new vector is still nondecreasing and differs by at most η in supremum norm, so the change of $\mathcal{T}X$ is at most $K(C)\eta$, regardless of the number of blocks. In a block with common parameter s , the tower property combines its transforms into a single conditional operation:

$$e^{sY_{b_j}} = \mathbb{E}[e^{sY_{c_j}} \mid \mathcal{F}_{b_j}].$$

Second, fix initial data and suppose that a random variable z and a random vector u are independent under the resulting conditional law. For $V(z, u) \in [-C, C]$ and $0 < m < m_0 \leq 1$, independence gives

$$T_m^{(z,u)} V = T_m^z (T_m^u V), \quad 0 \leq T_m^{(z,u)} V - \mathbb{E}_z [T_m^u V] \leq \frac{1}{2} C^2 e^{2C} m_0.$$

Here the superscript specifies the variables integrated. The inner quantity $T_m^u V$ lies in $[-C, C]$, so the inequality is precisely Equation (6.2) applied to the outer operation. It replaces one small-parameter logarithmic average over z by ordinary expectation while retaining the logarithmic average over u .

For either endpoint of a packet with D clauses, the leaf loss has absolute value at most $\log 2 + \beta D$ by Equation (5.7). With $D \leq D_0$, both consequences therefore apply with $C = \log 2 + \beta D_0$. They compare one packet at each endpoint. The removal argument will reduce a long horizon to one packet before invoking them.

7 A structural transition for shared variables

The trials with large horizons in Lemma 5.5 can use variables shared across sites. Our immediate objective is to reorganize one layer of that shared randomness while preserving the normalized trial bound at a fixed finite horizon. The records of conditional laws below quantify the information needed for the later replacement by independent site sampling.

7.1 Continuation laws retained at selected levels

Lemma 6.1 controls the cost of moving nearby averaging exponents. Once the exponents on a block have been made equal, its transforms combine into one conditional logarithmic average. We seek a probability measure at the block's start from which each fresh site's continuation can be sampled, conditionally independently of the other sites. The measures may still depend on the common ancestor data. Section 8 will choose finite test families fine enough to control the error of this replacement.

Fix an exponent width $\eta \in (0, 1)$. In an *active* hierarchy, choose boundary indices

$$1 = b_0 < \cdots < b_v = d.$$

The first level is reserved for the next reduction, and the consecutive blocks $(b_{i-1}, b_i]$, $1 \leq i \leq v$, are called processed. The case $v = 0$ is allowed when $d = 1$. In a *fully processed* hierarchy, the boundaries instead begin at $b_0 = 0$, so these blocks cover every level from 1 to d . Each processed block $(b, c]$ must satisfy

$$t_c - \eta \leq t_\ell \leq t_c \quad (b < \ell \leq c). \quad (7.1)$$

For a concrete example, consider the last processed block $(b, d]$. Fix a complete global path $z = (z_0, \dots, z_d)$ and a single site's history $u_{0:b} = (u_0, \dots, u_b)$. Continuing only that site's uniform sampling gives the message-pair law

$$\nu_{z, u_{0:b}} = \text{Law}_{u_{b+1:d}}(f^+, f^-) \quad \text{on } [0, 1]^2.$$

At boundary b we store a probability measure S_b that uses only the global and site history through b . We ask it to predict finitely many continuous test means of $\nu_{z, u_{0:b}}$, with a small mean-square error over $u_{0:b}$ for almost every fixed z . Thus the estimate averages ancestor site variables but does not average away dependence on later global variables. Earlier blocks require the same construction for the law of a probability-valued object, which leads to the iterated spaces below.

Let $K^{(0)} = [0, 1]^2$ with the sup metric and, recursively, let

$$K^{(j+1)} = \mathcal{P}(K^{(j)})$$

with the Wasserstein-1 metric induced by the preceding metric. These spaces are compact and have diameter at most one. At the bottom boundary put $S_d = (f^+, f^-) \in K^{(0)}$. Working upward through

the boundaries, if S_c has values in $K^{(j)}$, assign to S_b values in $K^{(j+1)}$. Each S_b is a measurable function only of the globals and the single site's generators through level b . A type is needed only at a boundary. The assigned type need not equal a conditional law exactly.

Choose finite nonempty families Φ_j of continuous functions $K^{(j)} \rightarrow [0, 1]$ and a tolerance $\delta > 0$. For a processed block $(b, c]$ with $S_c \in K^{(j)}$, the required record is

$$\int \left| \int \phi(S_c) \, du_{b+1:c} - S_b(\phi) \right|^2 \, du_{0:b} \leq \delta^2 \quad (\phi \in \Phi_j), \quad (7.2)$$

for almost every fixed complete global path, including its root. Here $u_{p:q}$ lists one site's generators $u_p, \dots, u_q$, all site integrals are Lebesgue integrals, and $S_b(\phi)$ is the integral of ϕ against the probability measure S_b .

The pathwise qualification preserves this record under the posterior of Lemma 5.6 and its additional positive first-level tilt. We keep the type functions unchanged, ignoring newly added global coordinates. Fresh site generators retain their product uniform law, and the enlarged global law is absolutely continuous with respect to the old law. The original exceptional set of complete global paths therefore remains null; the types are not recomputed as conditional expectations under the new posterior.

The depth-one reservoir trial starts active with boundary $1 = d$ and no processed blocks. The next transition adds one processed block while preserving every previous record. Its bound on inserted levels and its decrease of the first remaining exponent will force termination after finitely many reductions.

7.2 One step of the reduction

The following result turns a sequence of trials with growing horizons into a trial at a fixed horizon, with one more recorded block. Its input depth, boundaries, test functions, and output horizon are fixed before the input horizon tends to infinity.

Lemma 7.1 (Structural transition). *Fix a density $a > 0$, a finite penalty $\beta > 0$, $\eta \in (0, 1)$, $\delta > 0$, and $m_0 > 0$. Suppose there is a sequence of active trials of fixed depth d and fixed boundary indices, with horizons $N \rightarrow \infty$, satisfying*

$$\frac{G_N}{N} \leq U + o(1), \quad t_1 \geq m_0, \quad (7.3)$$

for a finite U . Suppose the old blocks satisfy Equations (7.1) and (7.2) with fixed finite continuous test families. Fix also a finite nonempty continuous $[0, 1]$ -valued test family on the type space of the current S_1 . After passing to a subsequence along which all old exponents converge, for each fixed positive integer k there is a finite-depth trial of horizon k with $G_k/k \leq U$ having the following properties.

For some $h \in \{0, 1, 2\}$, the output has depth $d + h$: the h new levels are inserted above the old first level, which becomes the sample level $h + 1$. Every old block $(b, c]$ becomes $(b + h, c + h]$ and retains its exponent band, test family, tolerance, and record. One new processed block ends at level $h + 1$ and satisfies Equations (7.1) and (7.2) for the specified new test family. Either this new block starts at the root, making the hierarchy fully processed, or it starts at the new first level and the hierarchy remains active, with first exponent

$$0 < m_{\text{new}} \leq (1 - \eta) \lim t_1. \quad (7.4)$$

All records hold for almost every complete global path.

The subsequent conditional-law and array subsequences may depend on the fixed output horizon k . The conclusion asserts no uniform rate as k varies; this is the order of limits used in the finite iteration later.

Proof strategy. In Subsections 7.3–7.4, we perturb the law at the initial level by a compensated Gaussian field. After selecting one conditional law, this perturbation gives identities for inner products of conditional site-test means. The identities remain valid when the test also observes all retained descendant and site data. Subsection 7.5 uses these identities to organize the samples into a nested partition at at most two thresholds. An exchangeability theorem then represents the partitioned array using independent variables along a sample tree and a site tree, including the variables shared by all sites.

Subsection 7.6 identifies the endpoint transforms by replacing each conditional integral temporarily with a finite average over independent children. This also determines the new exponents. Finally, in Subsection 7.7, empirical averages over fresh sites transfer closed inner-product constraints. A Hilbert-space support argument converts them into records for almost every complete shared path.

Throughout the proof the old depth, boundary indices, finite test families, and desired output horizon k are fixed. Write $m = t_1$ for the first input exponent. Along the input sequence $N \rightarrow \infty$, $m \geq m_0$, and Equation (7.3) holds. Pass to a subsequence on which all old exponents converge. We suppress the sequence index: before the weak limit below, m and t_j denote the input exponents; afterward they denote their limits. In both uses $m \geq m_0$. Expectations over the Gaussian fields below are denoted by $\mathbb{E}_g$. Their base probability measure is fixed before that expectation is taken.

7.3 Probing the first conditional site law

We first express the effect of the first shared variable on the chosen site tests. Fix the root data for the moment. Let $\Phi = \{\phi_1, \dots, \phi_{J_f}\}$, with $J_f \geq 1$, be the finite test family for the current first-boundary type S_1 . Root dependence is suppressed in the notation. Define

$$H_j(z_1, u_0) = \int \phi_j(S_1) du_1, \quad Q_j(z_1, z'_1) = \int H_j(z_1, u_0) H_j(z'_1, u_0) du_0. \quad (7.5)$$

These are Gram kernels of $[0, 1]$ -valued functions in the separable space $L^2([0, 1])$. Enumerate as (C_l) all nonconstant monomials in $Q_1, \dots, Q_{J_f}$, with nonnegative integer powers. Tensor products of their feature vectors show that these too are positive semidefinite, with diagonals at most one.

Each Q_j is an inner product of two conditional test means. We will derive identities for all these overlaps jointly, allowing the test also to observe the retained descendant data. This stronger test class will later be needed when we rank clusters by their masses. The next lemma supplies the identities uniformly over the conditional law of the first-level index.

The identities belong to the Ghirlanda–Guerra family of conditional overlap identities [10, arXiv version, Equation (28)]. The use of tests that also observe the retained descendant and site data follows the spin-tested formulation of Panchenko [25, Equation (14)]. We prove the conditional error estimate needed for the selected laws below.

Lemma 7.2 (Conditional Gaussian probing). *Let μ be a probability measure on a standard Borel space. Let $(C_l)_{l \geq 1}$ be positive semidefinite measurable kernels with separable Hilbert-space feature representations and $0 \leq C_l(\zeta, \zeta) \leq 1$. Put $a_l = 2^{-l}$ and let g_l be independent centered Gaussian fields with covariances C_l . For $s \geq 1$ and $x_l \in [1, 2]$, set*

$$h(\zeta) = s \sum_{l \geq 1} a_l x_l g_l(\zeta) - \frac{s^2}{2} \sum_{l \geq 1} a_l^2 x_l^2 C_l(\zeta, \zeta), \quad \mu^h(d\zeta) = \frac{e^{h(\zeta)} \mu(d\zeta)}{\mu(e^h)}. \quad (7.6)$$

The normalizer is finite and positive almost surely. Brackets denote conditionally independent samples from μ^h given the Gaussian fields, with any additional marks sampled by a kernel depending on those samples and with no further dependence on the fields. Define

$$V_l(\zeta) = \frac{g_l(\zeta)}{sa_l x_l} - C_l(\zeta, \zeta), \quad e_l = \mathbb{E}_g \left\langle \left| V_l(\zeta^1) - \mathbb{E}_g \langle V_l(\zeta^1) \rangle \right| \right\rangle.$$

Uniformly in μ , in the kernels, and in the other parameters,

$$\int_1^2 e_l \, dx_l = o_l(1) \quad (s \rightarrow \infty). \quad (7.7)$$

For every integer $n \geq 1$ and every bounded measurable f , $|f| \leq 1$, of the first n samples and their marks,

$$\left| n \mathbb{E}_g \langle f C_l(1, n+1) \rangle - \sum_{i=2}^n \mathbb{E}_g \langle f C_l(1, i) \rangle - \mathbb{E}_g \langle f \rangle \mathbb{E}_g \langle C_l(1, 2) \rangle \right| \leq e_l. \quad (7.8)$$

Here $C_l(i, j) = C_l(\zeta^i, \zeta^j)$.

Proof. Choose measurable feature maps v_j with $C_j(\zeta, \zeta') = \langle v_j(\zeta), v_j(\zeta') \rangle$; their norms are at most one. Fix l and write $x = x_l$. We allow x to range over $[0, 3]$ during the proof. Numerical constants denoted by K_l depend only on l . Consider

$$\Psi(x) = \frac{1}{s^2 a_l^2} \log \mu(e^h) + \frac{x^2}{2}, \quad A_x(\zeta) = \frac{g_l(\zeta)}{sa_l} - x C_l(\zeta, \zeta).$$

Differentiation gives

$$\Psi'(x) = \langle A_x \rangle + x, \quad \Psi''(x) = s^2 a_l^2 \langle (A_x - \langle A_x \rangle)^2 \rangle + 1 - \langle C_l(\zeta, \zeta) \rangle \geq 0. \quad (7.9)$$

The deterministic diagonal correction in (7.6) is exactly what permits both this convexity and the cancellation in (7.8) below.

For every fixed ζ , $\mathbb{E}_g e^{h(\zeta)} = 1$ and $-Cs^2 \leq \mathbb{E}_g h(\zeta) \leq 0$, with an absolute constant C . The two Jensen inequalities therefore give

$$-Cs^2 \leq \mathbb{E}_g \log \mu(e^h) \leq 0. \quad (7.10)$$

Consequently the convex function $\bar{\Psi} = \mathbb{E}_g \Psi$ is bounded on $[0, 3]$ by a constant depending only on l , and its one-sided derivatives are uniformly bounded on $[1/2, 5/2]$. Integrating (7.9) over $[1, 2]$ yields

$$\int_1^2 \mathbb{E}_g \langle (A_x - \langle A_x \rangle)^2 \rangle \, dx \leq K_l s^{-2}. \quad (7.11)$$

Interchanging expectation with the first derivative is legitimate here: on any compact interior interval, convexity bounds the derivative by secants to two fixed outer points, whose absolute values are integrable by (7.10) and Gaussian integrability.

The Gaussian gradient of $\log \mu(e^h)$ has norm at most Cs . Indeed it is the Gibbs mean of the direct-sum feature vector $(sa_j x_j v_j(\zeta))_j$, whose squared norm is at most $s^2 \sum_j a_j^2 x_j^2$. The Gaussian variance inequality gives

$$\mathbb{E}_g |\Psi(x) - \bar{\Psi}(x)| \leq K_l / s. \quad (7.12)$$

For $0 < u < 1/2$, compare $\Psi'(x)$ with its left and right secants of length u , and do the same for $\bar{\Psi}$. After integrating over $x \in [1, 2]$, (7.12) bounds the random secant errors by $K_l/(su)$. The remaining deterministic error is at most

$$\int_1^2 (\bar{\Psi}'_+(x+u) - \bar{\Psi}'_-(x-u)) \, dx \leq K_l u;$$

the integral telescopes and the interior derivatives are bounded. Thus

$$\int_1^2 \mathbb{E}_g |\Psi'(x) - \mathbb{E}_g \Psi'(x)| \, dx \leq K_l ((su)^{-1} + u). \quad (7.13)$$

Since $V_l = A_x/x$ and $x \geq 1$, the sum of (7.11), by Cauchy–Schwarz, and (7.13), with $u = s^{-1/2}$, proves (7.7).

For completeness, all the preceding operations can first be performed with finitely many Gaussian coordinates in finitely many feature spaces. The resulting bounds do not depend on these truncations. The Gaussian series converge in $L^2(\mathbb{P}_g \otimes \mu)$. For fixed s their limit is finite for μ -almost every index, for almost every field realization, so the normalizer is positive. Also $\mathbb{E}_g \mu(e^h) = 1$ by Fubini, so the normalizer is finite almost surely. For fixed s the exponentials have uniformly bounded moments of every fixed positive order, while Jensen’s inequality gives

$$\mu(e^h)^{-q} \leq \mu(e^{-qh}) \quad (q > 0).$$

The right side has uniformly bounded Gaussian expectation. These bounds give uniform integrability of the logarithms and of the replica integrals, including polynomial factors in g_l . They justify passage from finite-dimensional differentiation, concentration, and Gaussian integration by parts to the displayed formulas. Equivalently one can construct the fields by isonormal processes on the separable feature spaces; joint measurability is needed only outside a $\mathbb{P}_g \otimes \mu$ null set.

Average the additional marks conditional on their sample indices. The resulting test is independent of the fields, so Gaussian integration by parts gives

$$\mathbb{E}_g \left\langle f \frac{g_l(\zeta^1)}{s a_l x_l} \right\rangle = \mathbb{E}_g \left\langle f \left(\sum_{i=1}^n C_l(1, i) - n C_l(1, n+1) \right) \right\rangle.$$

Subtracting the diagonal term in V_l removes $C_l(1, 1)$. The case $f = 1$, $n = 1$ says $\mathbb{E}_g \langle V_l \rangle = -\mathbb{E}_g \langle C_l(1, 2) \rangle$. Finally,

$$|\mathbb{E}_g \langle f V_l \rangle - \mathbb{E}_g \langle f \rangle \mathbb{E}_g \langle V_l \rangle| \leq e_l,$$

which proves (7.8). □

7.4 Selecting one law and retaining its site data

For our concrete kernels the feature spaces are tensor powers of $L^2([0, 1])$. Fixed orthonormal bases in these spaces construct the Gaussian fields measurably also when the root global is random, before we condition on its value.

Apply Lemma 7.2 with $s = N^{1/4}$. Add $h(z_1)/m$ to both endpoint losses in the horizon- N comparison. This term passes unchanged through the transforms below the first level. The fixed- s exponential and inverse-normalizer bounds in the proof of Lemma 7.2 make the normalizer finite and positive almost surely and its logarithm integrable. Together with the conditional packet bound, these facts verify the integrability requirements for the added term in Lemma 5.6. Conditioning on all other endpoint data, its change to the first transform is

$$\frac{1}{m} \log \mu_*(e^h),$$

where μ_* is the unperturbed first-level posterior. By (7.10), the absolute change of its Gaussian-averaged value is at most Cs^2/m . The total perturbation error is therefore $o(N)$, uniformly while $m \geq m_0$.

Telescope the comparison in complete blocks of k packets, with a remainder of fewer than k packets, using Lemma 5.6. For a block, condition on all the base data belonging to the other packets, including their root-site variables and the original root global. Before the Gaussian tilt, its enlarged first-level law is then a fixed probability measure μ on an index ζ . This index includes the other packets' first-level variables. The kernels in (7.5) use its original z_1 projection only: the fresh site's variables remain independently uniform. All later posterior kernels depend on these base data and on their histories, but not otherwise on the Gaussian fields. Thus the marks used below have exactly the independence required in Lemma 7.2.

Choose a complete block uniformly. Its posterior k -packet functional, still averaged over Gaussian fields and fresh data, satisfies

$$\mathbb{E}_g G_k^{\text{post}} \leq kU + o(1) \quad (7.14)$$

on average over the block and base data and over independent uniform $x_l \in [1, 2]$. This follows from (7.3), the $o(N)$ endpoint error, and the $O_{a,\beta}(k)$ remainder bound. Furthermore, conditionally on any such base data and parameters,

$$|\mathbb{E}_g G_k^{\text{post}}| \leq C_{a,\beta} k. \quad (7.15)$$

Indeed a normalized logarithmic transform lies between the extrema of its argument, and each fresh packet has the root bound $|X| \leq \log 2 + \beta D$.

The uniform bounds (7.7), Markov's inequality, and a diagonal choice yield sets of choices of block, base data, and parameters of probability tending to one on which $e_l \rightarrow 0$ for every fixed l . More explicitly, choose a number of initial kernels tending to infinity so slowly that their summed mean errors tend to zero, and require that their sum of errors be at most the square root of that mean. Discard also the conditioning null sets on which a structural record fails. By (7.15), removal of the discarded set changes the mean in (7.14) by $o(1)$. For each N we may consequently choose deterministic block, base data, and parameters so that both (7.14) and all these vanishing-error conditions hold. The Gaussian fields remain averaged.

Under each selected conditional law, keep the Gaussian fields random. Conditional on them, sample countably many first-level indices independently from μ^h , in their natural order $u = 1, 2, \dots$. A natural label path has the form

$$(u, \xi), \quad u \in \mathbb{N}, \quad \xi \in \mathbb{N}^{d-1},$$

where ξ indexes the old descendant hierarchy below the first-level sample; for $d = 1$, it is the empty label. Draw independent fresh root-site uniforms for the sites $i \in \mathbb{N}$, with each site's root uniform shared by all its label paths. Conditional on the selected base and Gaussian fields, draw first-level site uniforms independently across sites and first-level sample indices. Below each sample, continue the old descendant hierarchy using its conditional global kernels and independent site uniforms. At each branch, distinct children use independent generators conditional on the parent history. Each site's root uniform is reused on all label paths; its later site uniforms are reused along their common label prefixes.

Let $\mathcal{B}$ be the old boundary set, including 1 and d . Retain at each path and site the decoration

$$D_{u,\xi,i} = (S_b^{u,\xi,i})_{b \in \mathcal{B}},$$

repeating each boundary value on its descendants. Also retain all pair kernels $(Q_j(u, v))$. Let $\mathcal{L}_N$ be the joint law of these countable arrays. This law averages the Gaussian fields and all the newly sampled data; the selected block, old-packet base data, and perturbation parameters are fixed.

Each decoration coordinate lies in one of the compact type spaces used in Equation (7.2), and every kernel coordinate lies in $[0, 1]$. The countable product is compact and metrizable, so along a subsequence $\mathcal{L}_N$ converges weakly to a law $\mathcal{L}$. The product term in (7.8) is a product of expectations under $\mathcal{L}_N$ and therefore converges to the corresponding product under this one limiting law. Write $\mathbb{E}$ for expectation under $\mathcal{L}$ in the identities below.

For $n \geq 1$, let $\mathcal{A}_n$ be the sigma-field generated by $D_{u,\xi,i}$ for $u \leq n$ and every ξ, i , together with $Q_j(u, v)$ for $u, v \leq n$ and $1 \leq j \leq J_f$. For every bounded $\mathcal{A}_n$ -measurable f and every bounded Borel function ψ on $[0, 1]^{J_f}$, the limit satisfies

$$n\mathbb{E}[f\psi(\mathbf{Q}(1, n+1))] = \mathbb{E}[f]\mathbb{E}[\psi(\mathbf{Q}(1, 2))] + \sum_{i=2}^n \mathbb{E}[f\psi(\mathbf{Q}(1, i))], \quad \mathbf{Q} = (Q_1, \dots, Q_{J_f}). \quad (7.16)$$

Indeed, first take f to be a continuous cylinder function of those decorations and kernels. It is an allowable mark-tested function in Lemma 7.2. Weak convergence gives the identity for nonconstant monomials ψ ; constants satisfy it directly. Polynomial approximation gives it for continuous ψ . Monotone-class arguments extend it first to bounded Borel ψ and then to every bounded $\mathcal{A}_n$ -measurable f . The joint law of the decorations and pair kernels is invariant under simultaneous permutations of the first-level sample indices. The hierarchical permutation invariances of the old descendant trees and the simultaneous permutation invariance of the sites also pass to the limit.

7.5 Ordered overlaps and marked finite cuts

We next turn the identities into a nested partition of the samples. The external ultrametricity theorem is stated for actual inner products of Hilbert-space samples. The following observation records the passage from our arrays to that setting, including the diagonal.

Lemma 7.3 (Ultrametricity from off-diagonal identities). *Let $R = (R_{ij})_{i,j \geq 1}$ be a symmetric, weakly exchangeable positive semidefinite array with $R_{ii} = 1$. Suppose that for every $n \geq 2$, every bounded Borel function f of the off-diagonal entries among the first n indices, and every bounded Borel function ψ , one has*

$$n\mathbb{E}[f\psi(R_{1,n+1})] = \mathbb{E}[f]\mathbb{E}[\psi(R_{12})] + \sum_{i=2}^n \mathbb{E}[f\psi(R_{1i})].$$

Then $R_{12} \geq \min\{R_{13}, R_{23}\}$ almost surely, and the analogous inequality holds for every triple of distinct indices.

Proof. The Dovbysh–Sudakov representation, in Proposition 1 of [22], realizes the array as a mixture of arrays

$$R_{ij} = \langle v_i, v_j \rangle + a_i \mathbf{1}_{\{i=j\}}, \quad a_i \geq 0,$$

where the pairs (v_i, a_i) are conditionally independent with the same law on a separable Hilbert space times $[0, \infty)$. The unit diagonal puts every v_i in the unit ball. Let $\mathcal{G}$ denote the random directing law of the vectors and put $q_* = \max \text{supp } \text{Law}(R_{12})$; this support is compact because positive semidefiniteness and the unit diagonal imply $|R_{ij}| \leq 1$.

Theorem 2(a) of Panchenko [21] applies to these off-diagonal Ghirlanda–Guerra identities and gives

$$\mathcal{G}\{v : \|v\|^2 = q_*\} = 1 \quad \text{almost surely.}$$

Its sphere conclusion has no finite-support hypothesis. Now form the true Gram array $\widehat{R}_{ij} = \langle v_i, v_j \rangle$, including its diagonal. It has the same off-diagonal entries as R and the deterministic diagonal q_* . A bounded test of its complete first- n matrix is therefore a bounded test of the off-diagonal matrix with constants substituted on the diagonal. Thus the displayed identities give the full Ghirlanda–Guerra identities for $\widehat{R}$ under the joint, averaged law. No conditioning on a realization of $\mathcal{G}$ is used in this implication.

The random measure $\mathcal{G}$ is supported on the unit ball of a separable Hilbert space. Theorem 1 of Panchenko [23] now applies to its true Gram array and gives the asserted ultrametric inequalities for the off-diagonal entries. This theorem likewise permits a general overlap distribution. $\square$

Lemma 7.4 (Ordering of the vector overlaps). *The support $\mathcal{S}$ of the off-diagonal law of $\mathbf{Q}$ in (7.16) is totally ordered componentwise. The scalar off-diagonal overlaps*

$$Q(u, v) = J_f^{-1} \sum_{j=1}^{J_f} Q_j(u, v), \quad u \neq v,$$

satisfy the ultrametric inequalities on triples of distinct indices, and, for $v, v' \in \mathcal{S}$,

$$|v_j - v'_j| \leq J_f \left| J_f^{-1} \sum_i (v_i - v'_i) \right|. \quad (7.17)$$

Proof. In each replica array, set $Q_j(u, u) = 1$ for every sample index u . Leave all $Q_j(u, v)$ with $u \neq v$ unchanged, even when the sampled values of z_1 coincide. Before passage to the limit this adds a nonnegative diagonal matrix to a Gram matrix, so it preserves positive semidefiniteness. The finite-matrix condition is closed and hence also holds in the limit. Every convex combination with positive rational coefficients is therefore a weakly exchangeable positive semidefinite array with unit diagonal. Its off-diagonal identities follow from (7.16), so Lemma 7.3 makes it ultrametric.

There are countably many rational combinations, so their conclusions hold simultaneously. In any off-diagonal triple the three vector overlaps must contain an equal pair. Otherwise some positive rational linear functional avoids their three equality hyperplanes and gives three distinct scalar overlaps, contrary to ultrametricity. If the repeated vector is v and the remaining vector is v' , then $v \leq v'$ componentwise: otherwise a positive rational functional concentrated on a violating coordinate would make the unique value the strict minimum, again impossible.

It follows that $\mathbf{Q}(1, 2)$ and $\mathbf{Q}(1, 3)$ are almost surely comparable. If two points in $\mathcal{S}$ were incomparable, they would have incomparable neighborhoods A, B of positive pair-law probability. Apply (7.16) with $n = 2$, $f = \mathbf{1}_{\{\mathbf{Q}(1, 2) \in A\}}$, and $\psi = \mathbf{1}_B$. Since $A \cap B = \emptyset$, it gives probability $\mathbb{P}(\mathbf{Q} \in A)\mathbb{P}(\mathbf{Q} \in B)/2 > 0$ of seeing an incomparable pair, a contradiction. Componentwise ordering implies (7.17); ultrametricity of Q was already obtained from the equal-weight combination. $\square$

For the rest of this subsection, Q denotes a generic off-diagonal overlap, with the law of $Q(1, 2)$. Fix the exponent width $\eta \in (0, 1)$ and a number $\gamma > 0$ to be chosen below. Let

$$q = \min\{t : \mathbb{P}(Q \leq t) \geq 1 - \eta\}, \quad x_p = \mathbb{P}(Q < q) \leq 1 - \eta.$$

We choose $r > q$ so that

$$x_c = \mathbb{P}(Q < r) > x_p, \quad x_c \geq 1 - \eta, \quad \text{diam supp}(Q \mid q \leq Q < r) \leq \gamma. \quad (7.18)$$

Here the conditioning event has positive probability. If $\mathbb{P}(q \leq Q < q + \gamma) > 0$, take $r = q + \gamma$. Otherwise q is not an atom, $\mathbb{P}(Q \leq q) = 1 - \eta$, and there is a gap of length at least γ to its right.

Let q' be the least support point of the remaining mass in $[q + \gamma, \infty)$ and take $r = q' + \gamma$. Then the band $[q, r)$ has its actual support in $[q', q' + \gamma]$, and (7.18) follows. This also covers endpoint atoms: r may exceed the maximum overlap.

Insert the parent relation $Q(u, v) \geq q$ if $x_p > 0$, and the child relation $Q(u, v) \geq r$ if $x_c < 1$. Make each relation reflexive on the diagonal. Ultrametricity makes them nested equivalence relations. The proper cuts have parameters

$$0 < x_1 < \cdots < x_h < 1, \quad 0 \leq h \leq 2. \quad (7.19)$$

All inequalities at a cut are non-strict, so atoms are included without ambiguity. The partition records only membership in these two relations. The overlap law within a band may still have infinitely many support points; the later record estimate will use the support diameter in Equation (7.18). Figure 1 illustrates the two-cut case.

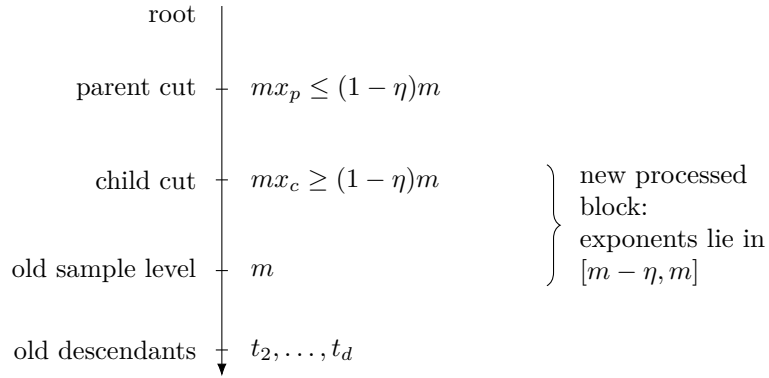


Figure 1: One structural transition when both cuts are proper. The parent exponent decreases by a fixed factor; the levels strictly below it through the old sample level form the new processed block. The cut parameters are $x_p = \mathbb{P}(Q < q)$ and $x_c = \mathbb{P}(Q < r)$. Trivial parent or child cuts are omitted, as specified in the proof. The old descendant row is absent when $d = 1$.

Panchenko's pure-state theorem proves hierarchical exchangeability of spin arrays and their independence from cluster weights under spin-tested Ghirlanda–Guerra identities [25, Theorem 1]. To describe the corresponding indexing here, rank cut clusters by decreasing frequency within their parents, resolving ties by the least natural sample index, and list samples within each deepest cluster in natural order. Write $(\alpha, l) \in \mathbb{N}^h \times \mathbb{N}$ for a sample's new address: α is its hidden-cluster address and l its number within that cluster. The decoration array becomes $D'_{\alpha, l, \xi, i}$. The whole old descendant subtree moves with its first-level sample u ; the site index i stays common. The following local argument proves independence of this reindexed array from the full partition that determines its addresses.

Lemma 7.5 (The marked cut representation). *The nested cut partition has the sample-partition law of an RPC with parameters (7.19). The reindexed decorations $D'_{\alpha, l, \xi, i}$ are independent of the full partition and are hierarchically exchangeable in the hidden-cluster and sample coordinates. They retain the independent hierarchical exchangeabilities of the old descendant trees and the simultaneous exchangeability of the sites.*

Proof. For a cut of parameter x_j , let C be a cluster seen among the first n samples, of size n_C . Select an anchor in C by a finite case distinction. Equation (7.16), with the indicator of the anchor's

cut relation as ψ , shows that, conditionally on all past pair and decoration data, the next sample belongs to C with probability

$$\frac{n_C - x_j}{n}. \quad (7.20)$$

Indeed the independent term contributes $(1 - x_j)/n$ and the past anchor-neighbor terms contribute $(n_C - 1)/n$. Differences between these probabilities specify the creation of new children within each parent. These are exactly the RPC predictions in Lemma 5.4, Equation (5.14). They determine every finite nested partition law. Consequently the full partition has that law, its clusters have positive frequencies, and the frequencies at every level have total mass one. The ranked leaf frequencies have the corresponding RPC distribution. For $h = 0$ these statements mean a single cluster of mass one.

We now prove independence from the full partition using the decoration tests in Equation (7.16). Let I be a deterministic ordered tuple of distinct sample indices, let D_I be all their decorations, and let Π be the full nested partition. Expose I first, using exchangeability. Given its cut pattern and decorations, iterating (7.20) shows that the conditional law of the partition on all remaining indices depends only on that cut pattern. In this iteration one can condition down from all intermediate decorations, since the predictions do not depend on them. Extending from finite restrictions to the countable partition gives, for every bounded decoration test f ,

$$\mathbb{E}[f(D_I) \mid \Pi] = K_{\Pi|_I}(f), \quad (7.21)$$

where the probability kernel K depends only on the ordered internal cut pattern $\Pi|_I$, and not on the actual values of the indices.

The inverse image of a fixed finite tuple of distinct positions in the ranked-cluster/natural-order indexing is a tuple of original indices measurable with respect to the partition. Partitioning according to its countably many possible values and applying Equation (7.21) proves that the reindexed tuple's decoration law is independent of the partition. Its internal cut pattern is fixed by its new positions. The same argument shows invariance under ancestry-preserving permutations of the hidden clusters and of the sample indices within them. It applies to arbitrary finite decoration cylinders and hence to the complete arrays. The old descendant permutations and the site permutations do not change the cut partition, so their invariances are retained as well. $\square$

We have now separated the RPC partition from the complete decoration array. The next representation supplies path variables for that array.

Combine the h hidden levels, the sample level, and the $d - 1$ old descendant levels into one label tree of depth $h + d$. Together with the depth-one site tree, the decorations form an array on a product of two finite-depth, countably branching trees. Their entry space, a finite product of compact type spaces, is standard Borel. The invariances just proved give the separate ancestry-preserving permutation invariances for these two trees. Theorem 2 of Austin and Panchenko [2] therefore represents this array by measurable functions of independent uniform variables indexed along the product of the two ancestor paths. Those paired with the site-tree root are shared globals z_l ; those paired with site i are site variables $u_{l,i}$. We take this representation independent of the partition, as permitted by Lemma 7.5. All shared globals are retained, including the root global. The representation concerns the site array alone; fresh packet counts and signs are still drawn independently at the root, as in Section 5.

An old type at a boundary shifted by h has a version using only the variables through that boundary. To see this, take two leaves agreeing through the boundary and diverging immediately afterward. Their boundary-type entries, for the same site, are equal almost surely by the duplicated-entry property. In the representation these two entries are conditionally independent and identically distributed given the prefix variables. Their conditional law is therefore a point mass. A measurable

version depends only on the prefix. There are only countably many entries, so these versions agree simultaneously throughout the array. At the last boundary there is no reduction to make.

7.6 Identifying the endpoint functional

The representation just obtained has new levels. We determine their exponents and their endpoint values from the original natural-order sampling. This calculation uses the full array law behind the finite partition, including the distribution within its bands.

Lemma 7.6 (Uniform empirical tree evaluation). *Let $C \geq 0$, let $B \geq 1$ be an integer, and let $|X| \leq C$ in a hierarchy of fixed depth d with exponents in $(0, 1]$. At every vertex replace its conditional integral by the empirical average over B conditionally independent children. The resulting nested logarithmic transform differs from the exact transform in mean absolute value by at most*

$$A(C, d)B^{-1/2},$$

uniformly in the conditional sampling kernels, the exponents, and the root data. The same statement holds for finitely many sites sampled jointly at every vertex.

Proof. For a one-step transform of a random variable $Y \in [-C, C]$,

$$T_t Y = \frac{1}{t} \log \mathbb{E} e^{tY}, \quad \hat{T}_t Y = \frac{1}{t} \log \left(\frac{1}{B} \sum_{b=1}^B e^{tY_b} \right).$$

The standard deviation of e^{tY} is at most $2tCe^C$, while $t^{-1} \log v$ is e^C/t -Lipschitz on $[e^{-tC}, e^{tC}]$. Hence

$$\mathbb{E} |\hat{T}_t Y - T_t Y| \leq 2Ce^{2C}/\sqrt{B}. \quad (7.22)$$

If exact child values Y are replaced by values Y' in $[-C, C]$, interpolation between the two sets of values gives

$$|T_t Y - T_t Y'| \leq e^{2C} \mathbb{E} |Y - Y'|,$$

and the identical bound with an empirical mean holds for an empirical transform. Apply these bounds recursively from the leaves upward. Every intermediate value remains in $[-C, C]$, so the constants depend only on C, d . A joint child variable may include the global coordinate and all the finitely many site coordinates; conditional independence between children is all that was used. $\square$

Lemma 7.7 (Endpoint passage). *Equip the represented hierarchy with exponents*

$$mx_1, \dots, mx_h, m, t_2, \dots, t_d. \quad (7.23)$$

For each endpoint of horizon k , its expected nested transform is the subsequential limit of the corresponding selected posterior endpoint. In particular its trial functional satisfies $G_k/k \leq U$. The hidden list is omitted when $h = 0$, and the old descendant list is omitted when $d = 1$.

Proof. First fix all packet counts and signs, so the endpoint uses finitely many distinct sites and its loss X is bounded, say by C . For each fixed B , the empirical tree evaluation is a continuous function of finitely many retained message values and of the old exponents. Continuity follows also at messages equal to zero or one, because every factor $1 - wf_1 f_2 f_3$ is at least $1 - w = e^{-\beta} > 0$. The old exponents converge to positive limits since $t_j \geq m_0$. Thus weak convergence of the natural-order arrays gives convergence of the expected empirical evaluation. Lemma 7.6 is uniform before the limit, so taking $B \rightarrow \infty$ afterward recovers exactly the subsequential expected endpoint value.

We now evaluate that same limit using the representation and its independent cut partition. Below the sample level, replace empirical evaluations by the true conditional transforms $T_{t_2}, \dots, T_{t_d}$. The error tends to zero by the same uniform estimate, even after propagation through the top empirical average. Write $Y_{\alpha, l} \in [-C, C]$ for the resulting value at sample position $(\alpha, l) \in \mathbb{N}^h \times \mathbb{N}$. Condition on the full cut partition and on all represented variables through the hidden levels, including their root variables. The remaining sample-level variables are independent across sample positions, and their laws are identical within each hidden leaf. If $\alpha(u)$ is the leaf containing natural sample u , then the conditional mean of the first- B natural-order average is

$$\frac{1}{B} \sum_{u=1}^B \mathbb{E}_{\text{sample}} e^{mY_{\alpha(u), 1}}.$$

Its conditional variance is $O_C(B^{-1})$. The empirical frequencies of the hidden leaves converge to their masses (v_α) ; convergence is in ℓ^1 , since these are probability masses whose pointwise limits have sum one. Hence, in L^1 ,

$$\frac{1}{B} \sum_{u=1}^B e^{mY_{\alpha(u), l(u)}} \longrightarrow \sum_{\alpha} v_{\alpha} \mathbb{E}_{\text{sample}} e^{mY_{\alpha, 1}}. \quad (7.24)$$

Boundedness makes passage to the divided logarithm legitimate. For $h = 0$ the sum consists of one term of weight one.

Set $V_{\alpha} = T_m^{\text{sample}} Y_{\alpha, 1}$. The represented marks and their root variables were chosen independently of the full partition, so the RPC weights retain their original law conditional on those root variables and are independent of the attached marks. If $h = 0$, the divided logarithm in (7.24) is simply $V_{\emptyset}$, where $\emptyset$ is the unique empty label. If $h \geq 1$, the hidden parameters satisfy $0 < x_1 < \dots < x_h < 1$, exactly the strict cascade hypothesis of Lemma 5.4. Applying its logarithmic identity conditionally on the root data to marks mV_{α} shows that the expectation of the divided logarithm on the right of (7.24) is the nested transform with hidden exponents $mx_1, \dots, mx_h$, followed by the sample exponent m . The scaling is the identity

$$T_x(mV) = mT_{mx}V,$$

applied successively at the hidden levels and followed by division by m . Below them remain the old exponents $t_2, \dots, t_d$. This proves precisely (7.23). The hidden exponents, when present, are increasing and lie strictly between zero and m . Every old descendant exponent is at least m , and their old order persists. Thus these are admissible intermediate trial exponents, including when some old exponents coincide or equal one.

Finally remove the conditioning on counts and signs. Both the prelimit and limit transforms are bounded in absolute value by $k \log 2 + \beta \sum D_i$, with the fixed Poisson packet laws. Their tails are uniformly integrable. Truncating counts, applying the proved finite-count assertion, and then removing the truncation establishes the assertion for each expected endpoint. Subtracting the B endpoint from the A endpoint and using (7.14) yields $G_k/k \leq U$. $\square$

7.7 Passing the pathwise structural records

The endpoint calculation has supplied a trial with the required value bound. It remains to preserve the old records and to construct the new one. We will pass empirical products of site tests through the weak limit; the following Hilbert-space fact then converts those pairwise bounds into statements about almost every individual global path.

Lemma 7.8 (Closed support bounds). *Let V, V' be independent random variables with the same law in a separable Hilbert space, and let $\delta, \varepsilon \geq 0$.*

1. *If $|\langle V, V' \rangle| \leq \delta^2$ almost surely, then $\|V\| \leq \delta$ almost surely.*
2. *If $\langle V, V' \rangle$ almost surely belongs to a fixed closed interval of length ε , then the support of their common law has diameter at most $\sqrt{2\varepsilon}$. Consequently $\|V - \mathbb{E}V\| \leq \sqrt{2\varepsilon}$ almost surely whenever the expectation exists.*

If V, V' are conditionally independent with the same conditional law given a standard Borel random element, the corresponding conclusions hold for almost every conditional law in which the stated pair bound holds.

Proof. A closed set of full product measure contains the product of the two supports: otherwise a pair of support points has product neighborhoods of positive measure disjoint from that set. The first assertion follows by putting the same support point in both coordinates. In the second, for support points v, v' , both squared norms and their inner product belong to the same interval, so

$$\|v - v'\|^2 = \|v\|^2 + \|v'\|^2 - 2\langle v, v' \rangle \leq 2\varepsilon.$$

Jensen's inequality gives the assertion about the barycenter. The same reasoning applies after disintegration; separability ensures that the relevant supports have full measure. $\square$

Lemma 7.9 (Preservation of the old records). *Every old processed block satisfies Equation (7.2) in the represented hierarchy, with the same test family and tolerance and with its boundaries shifted by h .*

Proof. Let $(b, c]$ be an old block. Since the input hierarchy is active, $b \geq 1$. In the prelimit natural array we may therefore choose two paths in the same first-level sample, agreeing through b and diverging immediately afterward. At site i , form the product of their two values of

$$\phi(S_c) - S_b(\phi), \quad \phi \in \Phi_j.$$

Condition on the complete globals of the two paths. The sites are independent and identically distributed, and the conditional mean of this product is the $L^2(du_{0:b})$ inner product of the two error functions

$$E_z(u_{0:b}) = \int \phi(S_c) du_{b+1:c} - S_b(\phi).$$

The fresh site variables after the branching point are independent on the two paths; their shared site variables are exactly those through b . Equation (7.2), which survived the posterior and Gaussian changes of law, bounds both error norms by δ . Cauchy–Schwarz therefore bounds the absolute conditional mean by δ^2 .

Let M_I be the empirical mean over the first I sites. The summands belong to $[-1, 1]$, so

$$\mathbb{E} \text{dist}(M_I, [-\delta^2, \delta^2])^2 \leq I^{-1}. \tag{7.25}$$

For fixed I , the integrand on its left is a bounded continuous function of finitely many retained type values: ϕ is continuous, and evaluation of a probability measure against ϕ is continuous in the type-space topology. The inequality thus passes to the limit. On $I = 2^j$, Markov's inequality and Borel–Cantelli imply that the displayed distance tends to zero almost surely. This holds simultaneously for all countably many path choices and all the finitely many tests. It consequently remains true for the partition-selected paths after reindexing.

In the product-tree representation, the conditional strong law for the sites identifies the limit of M_I with the inner product of the analogous error vectors, now in $L^2(du_{0:b+h})$. Conditional on the common globals through the shifted block start, those two vectors are independent and identically distributed as their continuation globals vary. Thus their inner product has absolute value at most δ^2 almost surely. Apply the first assertion of Lemma 7.8 to this conditional law. Each error vector has norm at most δ for almost every global continuation. Fubini's theorem gives precisely Equation (7.2) for almost every complete global path. The weak-limit step used a closed finite-column condition; the pathwise conclusion then followed from the conditional support bound. $\square$

Thus every previously processed block retains its record. For the new block, the narrow support band will provide the corresponding pairwise bound without a preexisting record.

Lemma 7.10 (A new processed block). *Let $c = h + 1$ be the sample level. Let $b = 1$ if the parent cut is present, and let $b = 0$ otherwise. There is an ancestor-measurable probability-valued type S_b such that $(b, c]$ satisfies Equation (7.2) for the specified new test family and tolerance δ . Its exponents lie in $[m - \eta, m]$. If the parent cut is present, its exponent belongs to $(0, (1 - \eta)m]$.*

Proof. For two distinct natural first-level samples, average over the first I sites the product of their $\phi_j(S_1)$ values. Conditional on their globals and Gaussian fields, the site's root variable is shared between the two samples, and its first-level variables are independent. The conditional mean is exactly Q_j from (7.5). Therefore

$$\mathbb{E} \left| \frac{1}{I} \sum_{i=1}^I \phi_j(S_1^{1,i}) \phi_j(S_1^{2,i}) - Q_j(1, 2) \right|^2 \leq I^{-1}. \quad (7.26)$$

For each fixed I this inequality passes by continuity to the natural-order limit array. Along dyadic I , it gives almost-sure convergence of the empirical average to Q_j , simultaneously for all pairs and tests.

After reindexing, take two sample positions agreeing through b and diverging immediately below b . If the parent is present, they are in the same parent cluster; otherwise $\mathbb{P}(Q < q) = 0$. If the child is present, they lie in distinct child clusters; otherwise $\mathbb{P}(Q < r) = 1$. Consequently their natural pair overlap satisfies $q \leq Q < r$ almost surely in all cases. By (7.18) and Lemma 7.4, the corresponding Q_j belongs to a fixed closed interval I_j of length at most $J_f \gamma$. The empirical characterization just proved shows that this interval condition is a property of the reindexed decorations alone; we do not need a separate representation of the kernels Q_j .

In the product-tree representation, the site empirical limit is the inner product in $L^2(du_{0:b})$ of the two vectors

$$A_j(u_{0:b}) = \int \phi_j(S_c) du_{b+1:c}. \quad (7.27)$$

Conditional on the ancestor globals through b , these are independent copies as their continuation globals vary. Their inner product belongs to I_j almost surely. The second assertion of Lemma 7.8 shows that the conditional support of A_j has diameter at most $\sqrt{2J_f \gamma}$.

Define, for each ancestor global and site history,

$$S_b(z_{0:b}, u_{0:b}) = \text{Law}_{z_{b+1:c}, u_{b+1:c}}(S_c). \quad (7.28)$$

All continuation variables in this formula have their current independent uniform laws. This is a measurable probability kernel on the type space of S_c , hence has exactly the required next type

space. By Fubini's theorem, $S_b(\phi_j)$ is the conditional continuation-global barycenter of the vector A_j in (7.27). The support diameter bound gives

$$\int \left| \int \phi_j(S_c) \, du_{b+1:c} - S_b(\phi_j) \right|^2 \, du_{0:b} \leq 2J_f \gamma$$

for almost every complete global path. Choose $0 < \gamma \leq \delta^2/(2J_f)$ to obtain the new record.

It remains to check the exponent band. The only possible intermediate level between b and c is the child cut, whose exponent is mx_c . By (7.18), $x_c \geq 1 - \eta$, so $m - mx_c \leq m\eta \leq \eta$. The end exponent is m . If the parent is present, its exponent satisfies $0 < mx_p \leq (1 - \eta)m$, as claimed.

The type in (7.28) is defined once, at this transition. In subsequent posterior changes it is retained as the same measurable function; it is not redefined by averaging under the changed global law. The pathwise record just proved then survives by absolute continuity, exactly as do the old records. $\square$

Completion of the proof of Lemma 7.1. The compactness and representation construction inserts at most two levels. Lemma 7.7 supplies the limiting expected endpoint values and $G_k/k \leq U$. Lemma 7.9 preserves every old record, while their exponent-band inequalities survive ordinary limits. Lemma 7.10 supplies the new processed block. If there is no parent cut it begins at the root and processing is complete. Otherwise the hierarchy is active, its first exponent is $mx_p > 0$, and this is at most $(1 - \eta)m$. All records hold for almost every complete global path. These constructions were made for the fixed output horizon k ; repeating them for any other fixed horizon proves the asserted quantifiers. All nonroot shared globals remain in this output. The next section removes them by a separate approximation argument. $\square$

8 Removing the shared variables and completing the trials

The transition in Lemma 7.1 keeps the shared variables and preserves the conditional records. We now use it finitely many times, then replace the nonroot shared variables by independent site sampling. The order matters: the replacement estimate is for one packet, so we first reduce the trial to a one-packet horizon. The final step approximates the resulting conditional sampling laws by finite rational tables.

Panchenko's theorem for finite replica symmetry breaking [26, arXiv version, Theorem 2] removes common nonroot coordinates while retaining common root data, under spin-tested Ghirlanda–Guerra identities, cavity equations for a modified Hamiltonian, and an overlap law with finitely many exact values. Here the overlap cuts partition a possibly continuous distribution. The recorded conditional laws and the quantitative estimates below provide the approximation needed to construct finite rational trials.

Panchenko's spin-distribution formula optimizes over a wider invariant distributional class in its stated even-arity setting; its shared-coordinate obstruction is explicit after Lemma 6 [24, arXiv version, Theorem 2 and the Remark after Lemma 6].

8.1 From recorded tests to probability kernels

Write W_1 for the Wasserstein metric associated with the metric on a compact type space. Every iterated type space $K^{(j)}$ in the structural record (7.2) is compact and has diameter at most one. Write $z = (z_0, \dots, z_d)$ for a complete global path.

Lemma 8.1 (Finite tests for the site law). *Given $e_0 > 0$ and a finite nonempty list of type spaces, one can choose finite nonempty collections Φ_j of $[0, 1]$ -valued 1-Lipschitz functions such that, for probability measures ν, S on $K^{(j)}$,*

$$W_1(\nu, S) \leq \max_{\phi \in \Phi_j} |\nu(\phi) - S(\phi)| + e_0/2. \quad (8.1)$$

If the structural record holds with

$$(\max_j |\Phi_j|) \delta \leq e_0/2, \quad (8.2)$$

then, for each processed block $(b, c]$ and almost every fixed global path,

$$\int W_1(\nu_{z, u_{0:b}}, S_b(z_{0:b}, u_{0:b})) \, du_{0:b} \leq e_0. \quad (8.3)$$

Here $\nu_{z, u_{0:b}}$ is the conditional law of S_c obtained by integrating only the fresh site coordinates $u_{b+1:c}$, with all the global coordinates fixed.

Proof. Kantorovich–Rubinstein duality expresses W_1 as the supremum of integral differences over 1-Lipschitz functions. Constants cancel from such differences; since the space has diameter at most one, each test can be shifted into $[0, 1]$. The class of these bounded Lipschitz functions is compact in the uniform norm. A finite $e_0/4$ -net gives Equation (8.1), since replacing a test by a net point changes the difference of its two integrals by at most $e_0/2$.

For a fixed test, the structural record and Cauchy–Schwarz give

$$\int |\nu_{z, u_{0:b}}(\phi) - S_b(z_{0:b}, u_{0:b})(\phi)| \, du_{0:b} \leq \delta.$$

Bound the maximum over tests by their sum, integrate Equation (8.1), and use Equation (8.2). This proves Equation (8.3) with its almost-everywhere global quantifier intact. $\square$

8.2 Finite stopping

Proposition 8.2 (A terminal one-packet trial). *Fix $U > P(a, \beta)$, $0 < \eta, m_0 < 1$, and an integer J such that*

$$(1 - \eta)^J < m_0. \quad (8.4)$$

Choose in advance finite nonempty test sets on $K^{(j)}$, $0 \leq j \leq J$, and a structural tolerance $\delta > 0$. For every $\rho > 0$ there is a trial with

$$G_1 \leq U + \rho,$$

at most J processed blocks, and depth at most $2J + 1$, which satisfies the chosen structural records and has one of the following forms:

1. *the processed blocks start at the root boundary 0;*
2. *the processed blocks start at boundary 1, and the remaining first exponent satisfies $0 < m < m_0$.*

Every processed block has exponent width at most η .

Proof. Lemma 5.5 supplies trials for arbitrarily large integer horizons N , initially of depth one and first exponent 1, with $G_N/N \leq U$. There are initially no processed blocks. A trial is terminal if its blocks start at the root or if its first unprocessed exponent is less than m_0 .

Suppose a sequence at a given stage contains no terminal trial. All its first exponents are then at least m_0 . Depth has increased by at most two at each preceding transition, so there are only

finitely many possible patterns of block boundaries. Pass to an unbounded horizon subsequence with a common pattern, and then to a subsequence on which the exponents converge. Lemma 7.1, with its fixed tests and tolerance, produces for each fixed positive integer k an output trial satisfying $G_k/k \leq U$. It preserves the old records and adds one processed block. Either the output is terminal, or its new first exponent is at most $(1 - \eta)$ times the limiting old first exponent.

This use of the transition involves nested limits, not a uniform limit in k . First fix k ; then take the input horizon to infinity as required by the transition; finally retain an output for that k . Doing this separately for $k = 1, 2, \dots$ gives the next unbounded horizon sequence if no terminal output occurs. After j transitions every still-active output has first exponent at most $(1 - \eta)^j$. Equation (8.4) therefore forces termination after at most J transitions. The resulting terminal trial has some finite horizon, at most J blocks, and depth at most $2J + 1$.

Reduce its horizon to one using the packet telescope of Lemma 5.6, without a Gaussian perturbation. There is no remainder when the blocks have size one. The sum of the averaged one-packet posterior differences is the original G_k , so one step, and then base root data outside the conditional null sets, have posterior value at most $U + \rho$. This selection is made before fresh packet data are sampled. The conditional laws generated by the posterior densities remain equivalent to the old laws, and the old site-law functions are kept rather than recomputed. Hence all the almost-everywhere records persist. Neither the exponents nor the terminal boundary changes. This gives the asserted one-packet trial. $\square$

8.3 Removing shared variables after stopping

Lemma 8.3 (Collapse by the recorded kernels). *Consider a terminal trial from Proposition 8.2, whose records imply Equation (8.3). Fix a count cutoff $D_0 \geq 1$ and put*

$$C = \log 2 + \beta D_0, \quad M = 3D_0, \quad R = JM e^\beta e^{2C(2J+2)} e_0.$$

There is a site-only trial, with weakly increasing exponents in $(0, 1]$, such that

$$|G_1^{\text{site}} - G_1| \leq 2K(C)\eta + 2R + C^2 e^{2C} m_0 + 2\tau_A + 2\tau_B, \quad (8.5)$$

where

$$\tau_A = \mathbb{E}[(\log 2 + \beta D_A) \mathbf{1}_{\{D_A > D_0\}}], \quad \tau_B = \mathbb{E}[(\log 2 + \beta D_B) \mathbf{1}_{\{D_B > D_0\}}],$$

with $D_A \sim \text{Poi}(3a)$ and $D_B \sim \text{Poi}(2a)$. In the root-terminal case the term containing m_0 is unnecessary. The new message kernels are independent of the fresh packet counts and signs.

Proof. We first condition on either endpoint's fresh count and sign data with $D \leq D_0$. Its loss X is bounded in absolute value by C , and it uses at most M distinct reservoir occurrences. Simultaneously replace all exponents in each processed block $(b, c]$ by t_c . The resulting vector is still nondecreasing and is within η of the original. Lemma 6.1 bounds the change by $K(C)\eta$. A block with constant exponent is one conditional T_{t_c} operation over all its coordinates, by the tower property.

Define deterministic functions of vectors of single-site types, one coordinate for each reservoir occurrence, working upwards through the block boundaries. At the last boundary put $F_d(\mathbf{S}_d) = X$. For a block $(b, c]$ put

$$F_b(\mathbf{s}) = \frac{1}{t_c} \log \int \exp(t_c F_c(\mathbf{u})) \prod_i s_i(du_i). \quad (8.6)$$

These functions depend on the fixed count and sign data, but the probability kernels S_b do not. The functions F_b are bounded in absolute value by C .

Use the sum metric on each finite product of type spaces. The terminal loss is e^β -Lipschitz: a clause factor is bounded below by $1 - w = e^{-\beta}$, the product of at most three $[0, 1]$ messages changes by at most the sum of their changes, and the log-sum-exp for the two branches of an A packet preserves the largest branchwise log error. Thus we may take $L_d = e^\beta$ as a Lipschitz constant for F_d . Coordinatewise couplings in Equation (8.6) give

$$L_b \leq e^{2C} L_c. \quad (8.7)$$

Indeed $v \mapsto e^{t_c v}$ has Lipschitz constant at most $t_c e^C$ on $[-C, C]$, and the divided logarithm on the possible integral values has Lipschitz constant at most e^C/t_c . The same two bounds show that a mean absolute error in an inside value propagates through a block by at most the factor e^{2C} .

Let Y_b denote the actual effective value at boundary b after flattening the processed blocks' exponents. Thus $Y_d = X$, and across a block $(b, c]$ the value Y_b is the conditional T_{t_c} average of Y_c over the block's global and site coordinates. Define

$$E_b = \mathbb{E}_{\text{base}} |Y_b - F_b(\mathbf{S}_b)|.$$

Here the expectation uses the current trial law on the global history and all occurrences' fresh site histories through b , before any exponential reweighting by this fresh packet. This law already includes any old-packet posterior used to obtain the trial; the packet count and signs remain fixed. In particular $E_d = 0$. We claim that

$$E_b \leq e^{2C} E_c + e^{2C} L_c M e_0. \quad (8.8)$$

First replace Y_c inside the block average by $F_c(\mathbf{S}_c)$. The mean-error propagation bound above gives the first term of Equation (8.8). To bound the remaining error, fix the ancestor histories through b and the continuation globals through c . Under this conditioning, the occurrences' fresh site continuations are independent; write ν_i for the resulting law of S_c^i . A product of Wasserstein couplings and the Lipschitz constant of F_c give

$$\left| \int e^{t_c F_c(\mathbf{u})} \prod_i \nu_i(du_i) - \int e^{t_c F_c(\mathbf{u})} \prod_i S_b^i(du_i) \right| \leq t_c e^C L_c \sum_i W_1(\nu_i, S_b^i).$$

The second integral depends only on the ancestor histories. Average this inequality over the continuation globals, apply the e^C/t_c Lipschitz bound for the divided logarithm, and then average over the ancestor histories. For almost every fixed global path, Equation (8.3) bounds the integral over each occurrence's site ancestors by e_0 . The continuation-global law is independent of these fresh site generators, including when the trial was obtained by a posterior change of law. Fubini's theorem therefore bounds the final average by $e^{2C} L_c M e_0$ and proves Equation (8.8).

There are at most J processed blocks, and $L_c \leq e^{2CJ} e^\beta$ at every boundary. Iterating Equation (8.8) from $E_d = 0$ bounds their accumulated error. If the starting boundary is 1, propagating once through the remaining first-level transform adds at most one factor e^{2C} . In either case the error is at most

$$J M e^\beta e^{2C(2J+2)} e_0 = R.$$

If the processed blocks begin at the root, the resulting recursion already describes a site-only trial. Retain the common root z_0 and sample each occurrence's initial type $S_0(z_0, u_0^i)$ using its own fresh root-site uniform. At one level of exponent t_c for each old block, draw each site's next type independently from its preceding type, and continue to the message pair. Probability kernels on these compact metric spaces can be sampled by measurable functions of fresh uniform variables. This construction evaluates Equation (8.6).

If the blocks begin at boundary 1, there is still the operation $T_m F_1(\mathbf{S}_1)$, where $0 < m < m_0$. Conditional on the root data, factor this operation as

$$T_m^{z_1} T_m^{\text{sites}} F_1(\mathbf{S}_1).$$

The inside quantity is in $[-C, C]$. By Equation (6.2), replacing the outside $T_m^{z_1}$ by ordinary expectation changes its value by at most $\frac{1}{2}C^2 e^{2C} m_0$. Move z_1 , with this ordinary sampling law, into the root global data. Retain the level-1 site transform and the independent block draws just constructed. This is again site-only. Its kernels were constructed from the recorded types alone, not from any fresh count or sign data.

For counts greater than D_0 , both the old and new endpoint transforms are bounded in absolute value by $\log 2 + \beta D$, regardless of their kernels or exponents. Their contribution to the difference is thus at most twice the corresponding tail expectation. Summing the bounded-count estimates for the two endpoints proves Equation (8.5). $\square$

8.4 Rational finite descriptions

Theorem 8.4 (Finite rational completeness). *For every $a > 0$ and finite $\beta > 0$,*

$$P(a, \beta) = \inf_{\mathcal{Q}} G_1, \tag{8.9}$$

where $\mathcal{Q}$ consists of site-only trials of finite depth whose exponents are strictly increasing rationals in $(0, 1)$ and whose message pair is a rational-valued step function on a finite rational rectangular grid in its root and site uniform variables. In particular, this is a countable class with an effective enumeration.

Proof. The upper bound of Proposition 5.3 gives $P(a, \beta) \leq G_1$ for every member of $\mathcal{Q}$. We prove the reverse inequality by giving the order of all approximation choices.

Fix $\epsilon > 0$ and put $U = P(a, \beta) + \epsilon$. First choose a count cutoff $D_0 \geq 1$ with $\tau_A, \tau_B \leq \epsilon$, and set $C = \log 2 + \beta D_0$. Choose $\eta, m_0 \in (0, 1)$ so that

$$K(C)\eta \leq \epsilon, \quad \frac{1}{2}C^2 e^{2C} m_0 \leq \epsilon.$$

Next choose J satisfying Equation (8.4), and only then choose $e_0 > 0$ so small that

$$J(3D_0)e^\beta e^{2C(2J+2)}e_0 \leq \epsilon.$$

Choose the finite test sets on all type spaces through $K^{(J)}$ by Lemma 8.1, and finally choose δ satisfying Equation (8.2). In particular, neither the test precision nor the allowed number of transitions is chosen in response to the eventual hierarchy depth. The overlap widths used inside each transition are chosen only after these test sets and δ , as in Lemma 7.1.

Proposition 8.2, with $\rho = \epsilon$, gives a terminal one-packet trial of value at most $P + 2\epsilon$. Lemma 8.3 turns it into a site-only trial whose value is at most $P + 12\epsilon$: the exponent changes contribute at most 2ϵ , the kernel replacements at most 2ϵ , the possible small-exponent replacement at most 2ϵ , and the two tails at most 4ϵ .

It remains to obtain a finite rational description. Approximate the finite weakly increasing exponent vector by strictly increasing rational vectors in $(0, 1)$. This is possible even if some original exponents coincide or equal 1. For fixed count data, Lemma 6.1 gives convergence of the transform values. The packet bound of Equation (5.7), uniform over the approximations, then gives convergence after averaging the Poisson counts and signs.

Encode the root randomness by a single uniform coordinate. After the independent conditional type samplings have likewise been encoded by uniform coordinates, the message pair is a measurable function of finitely many independent uniforms: the root global, the root site coordinate, and the finitely many site-level coordinates. Approximate this pair in L^1 by rational-valued step functions on finite dyadic rectangular grids. For example, conditional expectations on the successive dyadic grids converge in L^1 , and subsequent rational quantization preserves the range $[0, 1]^2$ and the convergence. The shared root coordinate is retained as shared; no independence across sites is introduced there.

For fixed counts and signs, the leaf Lipschitz estimate in the proof of Lemma 8.3 bounds the mean leaf error by e^β times the sum of the L^1 message errors over the finitely many occurrences. The mean-error propagation bound through the fixed finite hierarchy proves convergence of both endpoint values. Once more, the root linear-in-count bound makes the Poisson tails uniform, so their fully averaged G_1 values converge. Thus a member of $\mathcal{Q}$ has value at most $P + 13\epsilon$.

Since ϵ is arbitrary, Equation (8.9) follows. Finite depths, finite grids, rational exponents, and finite rational tables all have finite encodings and can be enumerated by a single ordinary algorithm. These are the trials used in the certificate search; none of the compactness limits or tolerance choices in the existence proof is input to that search. The larger class of strict measurable site-only trials contains $\mathcal{Q}$ and obeys the same upper bound, so its infimum also equals $P(a, \beta)$. $\square$

9 Completing the computation

We now have the two certificate families described in Section 2. Let α be the threshold constructed in Theorem 4.4. Proposition 4.5 gives a computable rational sequence (ℓ_n) with $\ell_n < \alpha$ and $\ell_n \rightarrow \alpha$. Its underlying finite-size bound is the exact one-sided estimate

$$\frac{f_n}{n} - 2^{67} n^{-1/6} \leq \alpha.$$

For the upper certificates, a rational trial is the finite site-only description of Section 5: it has strictly increasing rational exponents in $(0, 1)$ and a rational-valued message pair on a finite rational rectangular grid of the root and site uniform variables.

Lemma 9.1 (Upper certificates). *For positive rational a , positive integer β , and a rational trial Q , the inequality $G_1(a, \beta; Q) < 0$ certifies $\alpha \leq a$. Every rational $a > \alpha$ has such a certificate. The class of certified densities is effectively enumerable.*

Proof. The trial bound of Proposition 5.3 gives $P(a, \beta) \leq G_1(a, \beta; Q)$. If $a < \alpha$, the pressure exists by Lemma 5.2 and is nonnegative by Corollary 4.7. Thus a negative trial is impossible below α and certifies the non-strict inequality $\alpha \leq a$. Nothing about the pressure at equality is needed.

If $a > \alpha$, Corollary 4.7 supplies a positive integer β for which $P(a, \beta) < 0$. Theorem 8.4 says that $P(a, \beta)$ is the infimum of the rational trial values. A rational trial therefore has negative value. Its finite description is eventually enumerated, and Lemma 5.1 eventually gives a rational interval for its value with negative upper endpoint. Enumerating positive rational densities, positive integer penalties, finite descriptions and accuracy indices gives the asserted enumeration. $\square$

Proof of Theorem 1.1. Theorem 4.4 proves the strict-side probability limits for the proper three-clause model in the introduction, with $\alpha \in [1/200, 10]$. Set $\alpha_3 = \alpha$. This also identifies the constant with the same-model $k = 3$ threshold in [19, Theorem 1.1] by the uniqueness argument in Section 3.

Use the sequence from Proposition 4.5 as the lower sequence in Lemma 2.1, and use $G = G_1$ on the rational trial class. This class is effectively enumerable by its finite encoding, and Lemma 5.1 proves uniform computability of its values. Lemma 9.1 proves both sign hypotheses of the search

lemma. Since $\alpha \in [0, 20]$, all its assumptions hold. The resulting single finite machine halts on every unary input 1^r and returns the required binary-encoded rational within 2^{-r} .

The compactness subsequences, conditional laws, Gaussian parameters and approximation tolerances used to prove completeness are not inputs to the machine. They establish the existence of finite witnesses, which the search finds through its fixed enumeration. The running time of this fixed machine on 1^r is itself a total computable function of r , obtained by simulating the halting computation; the proof supplies no efficiency estimate for that function. $\square$

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